A Data-Driven, Component-Level Projection Framework for 21st-Century Sea-Level Rise

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Abstract

As the rise in global mean sea level (GMSL) accelerates and warming carries projections beyond the conditions their models were calibrated on, the central challenge is validating those models against withheld observations spanning a comparable change in warming. We present a framework for projecting GMSL that uses a simple physical model of each contributor, and we withhold observed GMSL from every calibration to test the sum of the components. Thermal expansion, glaciers, the Antarctic Peninsula, and Greenland ice discharge, modeled as a delayed response to ocean warming, are calibrated by Bayesian inference against independent observations. Greenland surface mass balance is emulated, and East Antarctic snowfall and terrestrial water storage are adopted from published work. The component sum accounts for 89–104% of the satellite-altimetry rate since 1993. Forced with observed warming over 1950–2025 (1.0–1.1 °C), the components match satellite-era rates but rise 0.3–0.5 $\mathrm{mm\,yr^{-1}}$ too slowly over 1950–2000. Whether West Antarctic Ice Sheet (WAIS) loss will accelerate is unconstrained by observations, so we use two storylines. Under the slow-WAIS storyline, which assigns 90% probability to continued loss along the observed trajectory, median GMSL rise over 2000–2100 is 0.45, 0.54, and 0.64 m for scenarios leading to 1.8, 2.7, and 3.6 °C above pre-industrial. Under the fast-WAIS storyline, which assigns 90% probability to accelerated loss, the medians are 0.59, 0.69, and 0.78 m. Its 95th percentile lies about 1 m above the median in all three scenarios, because we treat WAIS loss this century as independent of emissions, as recent studies suggest.

1 Introduction

Sea-level rise will continue for centuries even if warming stops, so the question for coastal communities is not whether to adapt but how fast (*Fox-Kemper et al.*, 2021). About 250 million people live on land below the present annual flood level, and the sea-level rise projected by 2100 under the RCP4.5 emissions scenario would raise that number to about 360 million, at 2010 population and without coastal defenses (*Kulp and Strauss*, 2019). Without further adaptation, global flood damages could reach US$14 trillion per year at 0.86 m of rise (*Jevrejeva et al.*, 2018). Coastal planners use widely varying projections (*Hirschfeld et al.*, 2023). The Sixth Assessment Report of the Intergovernmental Panel on Climate Change (AR6) projects median rise of 0.44 and 0.56 m by 2100, relative to 1995–2014, for low- and intermediate-emissions scenarios (SSP1-2.6 and SSP2-4.5) (*Fox-Kemper et al.*, 2021). AR6

also attributes deep uncertainty to ice-sheet processes that models do not represent well (*Fox-Kemper et al.*, 2021). High-end estimates published since AR6 span 0.6–2.0 m of rise by 2100 (*Garner et al.*, 2026).

Whatever the method behind these projections, continued warming ensures that they will remain out of sample with respect to the observational record. The central difficulty is therefore not building a model but establishing when its projections can be trusted. A model earns that trust when it is calibrated with observations and tested against observations withheld from calibration. Projections can also be tested against observations made after they were published. The mid-range projection of the IPCC Second Assessment Report matched the global sea-level rise of the following three decades to within about 10 mm, while its projections of individual contributors were more variable (*Törnqvist et al.*, 2025). A test against withheld data is most informative when the withheld record spans a change in forcing comparable to the change being forecast. Such a test does not remove the out-of-sample problem, but it estimates how far a model can be carried beyond the conditions it was calibrated on. A model must also report its uncertainties in a form that separates their sources (*Aschwanden et al.*, 2021). A recent community synthesis makes this case, calling for projections narrowed by observations and for uncertainty characterized by its structure and not only by its magnitude (*Felikson et al.*, 2025).

Sea-level projection models span a wide range of complexity, and they meet this standard to different degrees, but none fully. At the most detailed end are process-based ice-sheet models, which solve the equations of ice flow and deformation and can test competing hypotheses about the mechanics of ice loss (*Schoof*, 2007; *Morlighem et al.*, 2024). They are needed to project ice loss under conditions outside any observational record. Reproducing observed mass change is a necessary, though not sufficient, condition for trusting their projections (*Aschwanden et al.*, 2021), and this condition is hard to meet. One obstacle is that their parameters are inferred by solving inverse problems in which the data constrain only a few combinations of parameters (*Petra et al.*, 2014; *Isaac et al.*, 2015). As of 2020, no ice-sheet model run forward in time had reproduced the sixfold increase in ice-sheet mass imbalance observed since the early 1990s (*Siegert et al.*, 2020). Most of the recent Antarctic projections grow the ice sheet over the first few years of the simulations, while observations over the same years show it losing mass (*Fricker et al.*, 2025; *Seroussi et al.*, 2024). Projections also depend on how each model is initialized and calibrated (*Seroussi et al.*, 2020; *Goldberg et al.*, 2026).

Models run to equilibrium with past climate before the projection begins change little on their own but miss the observed present-day mass loss. Models initialized to match the present geometry and flow may not reproduce the observed rate of mass change (*Pattyn and Morlighem*, 2020; *van den Akker et al.*, 2025). In the Antarctic ensemble of the Ice Sheet Model Intercomparison Project (ISMIP6), the choice of ice-sheet model accounts for 52% of the uncertainty in projected dynamic mass loss by 2100, compared with 13% for the choice of climate model (*Seroussi et al.*, 2023), and the projected Antarctic contribution to 2100 shows no clear response to the emissions scenario (*Edwards et al.*, 2021). These models also lack a validated representation of iceberg calving and fracture, and many hold the calving front fixed (*Pattyn and Morlighem*, 2020; *Bassis et al.*, 2024; *Benn et al.*, 2026). AR6 complements these models with structured expert judgment in its low-confidence projections (*Bamber et al.*, 2019; *Fox-Kemper et al.*, 2021).

At the other extreme are approaches anchored directly in the observed record of global mean sea level (GMSL). The simplest extrapolates the observed rate and acceleration of GMSL rise forward in time (*Nerem et al.*, 2022). Semi-empirical models instead fit the rate of GMSL rise to global mean surface temperature (GMST) (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Rahmstorf et al.*, 2012; *Kopp et al.*, 2016). Both are transparent and fit the quantity being predicted directly. However, they give no insight into which processes drive the rise. The fitted relationship also need not hold when contributors with different response times change their relative importance (*Church et al.*, 2013).

Between these extremes sit component-level models, which constrain each physical contributor separately, trading spatial detail for parsimony and transparency (*Perrette et al.*, 2013; *Kopp et al.*, 2014; *Mengel et al.*, 2016; *Nauels et al.*, 2017; *Wong et al.*, 2017; *Kopp et al.*, 2023). Component-level frameworks differ in how each contributor is constrained and in whether the observed GMSL record is used in calibration or kept for testing. When every calibrated component is fit to its own independent record and GMSL is withheld throughout, as in *Mengel et al.* (2016), the sum of the calibrated components can be tested against observed GMSL.

No single class of model can be at once the most detailed in physical processes, the most transparent, and the most tightly constrained by data. Ordered by complexity, models can instead form a hierarchy in which each level is calibrated and tested against data before the next level of complexity is trusted. Ordering models from idealized to comprehensive is long established in climate science, where it has helped close the gap between simulation and understanding (*Held*, 2005; *Jeevanjee et al.*, 2017; *Maher et al.*, 2019). Such a hierarchy does not require any one model to be correct. It requires only that its models rest on assumptions different enough that their agreement or disagreement is informative. Where structurally different models agree, a result rests on more than one set of assumptions (*Lipscomb et al.*, 2025). Where they disagree, the divergence shows what the data leave unconstrained and which observations would constrain it.

Simpler levels also bring practical advantages. Across 97 published comparisons, simple forecasting methods were as accurate as or more accurate than complex ones in 84% of cases (*Green and Armstrong*, 2015). In a competition on 100,000 time series, about two-thirds of the submitted methods did not beat a simple combination of exponential-smoothing forecasts (*Makridakis et al.*, 2020). The assumptions of simple models are also easier to state and check, which helps those who use the results find errors and omissions (*Green and Armstrong*, 2015). A hierarchy built this way complements process-based ice-sheet models and aids their development rather than substituting for them.

Here we construct a data-driven, component-level forecast of 21st-century sea-level rise. We call the framework SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Each contributor that the observational record can constrain is represented by a parsimonious physical model and calibrated by Bayesian inference against its own independent record. Among these, Greenland ice discharge is modeled as a lagged response to the temperature of the surrounding ocean, calibrated with observed discharge. Observed GMSL is withheld from every calibration. The summed components can therefore be tested against observations from 1950 to 2025, a period spanning about 1.0–1.1 °C of warming, comparable to the warming that remains this century under intermediate emissions. We place SPECTRUM in a hierarchy and compare its forecast with

extrapolation of the observed trend, a semi-empirical model, and the AR6 assessment. Where the record cannot constrain a contributor, as for the West Antarctic Ice Sheet (WAIS), we say so and represent the range of outcomes with storylines rather than a calibrated model, keeping this source of uncertainty separate from the others. This separation lets us ask which part of 21st-century sea-level rise can be reduced by cutting emissions. It also shows which part of the uncertainty can be narrowed only by observations and models that constrain how fast West Antarctica will lose ice.

2 A data-driven, component-level projection framework

SPECTRUM represents each contributor to GMSL with its own model, and the way each model is constrained depends on the data available (Fig. 1). Thermal expansion, glaciers, the Antarctic Peninsula, and Greenland ice discharge are calibrated by Bayesian inference against independent multi-decade observational records, listed in Table S1 of the Supplementary Material. Greenland surface mass balance is emulated from regional climate model simulations. East Antarctic surface mass balance and terrestrial water storage are adopted from published sensitivities and assessments, and West Antarctica is represented by slow and fast storylines. Each temperature-sensitive component is forced by GMST, directly or through a regional temperature derived from it, and its temperature dependence, calibrated or adopted, defines its forecast. Observed GMSL is withheld from the calibrations and used to test the sum of the components.

Earlier component-level frameworks constrain their contributors in different ways. Several draw their components from process models, assessments, or expert judgment. *Kopp et al.* (2014) build the distribution of each contributor from process-model ensembles, the AR5 assessment, and expert elicitation. The MAGICC sea-level model (*Nauels et al.*, 2017) emulates process-based projections, with each component calibrated with output from climate, glacier, or ice-sheet models. FACTS (*Kopp et al.*, 2023) draws its land-ice contributions from emulators of model intercomparison projects or from structured expert judgment. Others use observed GMSL in calibration. BRICK (*Wong et al.*, 2017) calibrates its component models against observations and then conditions the assembled ensemble on observed GMSL. In the high ice-sheet case of *Perrette et al.* (2013), the ice-sheet contribution is the residual of a semi-empirical model calibrated with observed GMSL. The design closest to SPECTRUM is that of *Mengel et al.* (2016), who calibrate a semi-empirical relation for each contributor against its own observations, constrained by estimates of long-term commitment, and compare the sum with total sea-level reconstructions not used in the calibration. SPECTRUM follows that design and differs from it in three ways. Its parameters are estimated by Bayesian inference, which yields their joint posterior distribution, rather than by least squares. Greenland ice discharge is forced by the temperature of the ocean around Greenland, whereas *Mengel et al.* (2016) drive it with GMST. Finally, *Mengel et al.* (2016) note that their method cannot represent processes that proceed independently of the forcing, such as a collapse of the Amundsen Sea sector of West Antarctica. SPECTRUM represents the range of West Antarctic outcomes with storylines instead.

Unless otherwise noted, all sea-level values in the main text are global-mean 21st-century contributions referenced to the year 2000. Throughout, we label each emissions scenario by

the AR6 central estimate of its global mean surface temperature over 2081–2100 relative to the 1850–1900 (pre-industrial) mean (*Lee et al.*, 2021, Table 4.5): 1.4 °C (SSP1-1.9), 1.8 °C (SSP1-2.6), 2.7 °C (SSP2-4.5), 3.6 °C (SSP3-7.0), and 4.4 °C (SSP5-8.5). The central paths of the 1.8, 2.7, and 3.6 °C scenarios reach 1.7, 2.7, and 3.9 °C in 2099, the last year of the AR6 projections. We hold each path at its 2099 value thereafter.

2.1 Component models, calibration, and validation

2.1.1 Greenland Ice Sheet

The Greenland Ice Sheet (GrIS) is among the largest land-ice contributors to sea-level rise and has lost mass every year since 1998 (*Mouginot et al.*, 2019). That loss is the imbalance between solid-ice discharge and surface mass balance (SMB), where SMB is the difference between mass gained by snow accumulation and mass lost to meltwater runoff and sublimation. GrIS discharge is sensitive to ocean temperature and surface melt runoff (*Straneo et al.*, 2013; *Slater et al.*, 2020; *Wood et al.*, 2021). Warm subsurface waters thin and undercut the calving fronts of marine-terminating glaciers, driving increased calving and net terminus retreat. This effect is amplified where surface runoff drives mixing and buoyant upwelling along the calving face (*Slater and Straneo*, 2022). Surface melt also lubricates glacier beds, but that effect is generally short-lived and localized as the subglacial hydrological system evolves to accommodate changes in meltwater flux (*Schoof*, 2010; *Tedstone et al.*, 2013; *Ultee et al.*, 2022). We therefore model discharge as a lagged response to ocean temperature.

2.1.1.1 Discharge

We approximate the GrIS-wide discharge as

$$D = \gamma\, T_{\mathrm{ocean}}(t - \delta) + r_0, \tag{1}$$

where D is the GrIS discharge in sea-level equivalent (SLE) units, γ is an effective discharge sensitivity to ocean temperature, δ is the lag between warming of the ocean around Greenland and the response of discharge, r_0 is the background discharge rate at the baseline ocean temperature, and T_{ocean} is the ocean temperature anomaly in the waters around Greenland relative to the 1995–2005 mean. SLE is the rise in global mean sea level that would result if a given ice mass were added to the ocean, with 1 mm SLE = 362.5 Gt throughout this study. Because D is the gross discharge flux rather than an anomaly, it becomes a contribution to GMSL only when combined with SMB. We emulate SMB separately from a regional climate model (Section 2.2.1).

We calibrate against total GrIS discharge, summed over the 219 glaciers classed as tidewater in the *Mouginot et al.* (2019) dataset, of which 107 (85% of discharge) are measured directly. Summing gives up spatial detail but reduces noise relative to the signal. Our goal is to capture ice-sheet-scale discharge rather than that of individual basins and glaciers. The calibration is carried out on the time-integrated form of Eq. 1, which gives the cumulative discharge,

$$H_{\mathrm{D}}(t) = \int_{t_0}^{t} D(t')\,\mathrm{d}t', \tag{2}$$

with γ and r_0 held constant in time. Fitting the cumulative record reduces the influence of single-year noise and directly constrains the total mass lost, the quantity of interest in this study. Its errors are correlated in time, which we account for below.

The calibration uses the ice discharge time series of *Mouginot et al.* (2019), which spans the 47 years from 1972 to 2018, and ocean temperatures from EN4 (v4.2.2) (*Good et al.*, 2013). We average EN4 over 200–500 m depth, the range most relevant to the grounding lines of Greenland's tidewater glaciers (*Slater et al.*, 2020), and over the seas surrounding Greenland south of 80°N (58–80°N, 75–5°W). This geographic limit is set by the sparse subsurface sampling farther north. We use EN4 from 1970 onward, so the delayed forcing is undefined in the first years of the discharge record, and every candidate lag is fit to the same 41 years, 1978 to 2018, those covered by the longest lag.

Because the data are resolved annually, the lag δ takes integer-year values. For each candidate lag from 4 to 8 yr, we fit (γ, r_0) by weighted least squares to the annual increments of the cumulative discharge, which are the yearly discharge values, each weighted by its reported uncertainty (Fig. 2). The cumulative record is a running sum of these yearly values, whose errors are independent, so this fit accounts for the correlation of errors in the cumulative record. The residuals scatter more widely than the reported uncertainties allow, with reduced chi-square $\chi^2/\mathrm{dof} = 1.45$ on 38 degrees of freedom at the preferred lag, where χ^2 is the error-weighted sum of squared residuals. We therefore scale the reported uncertainties by a common factor estimated from the fit, which widens the parameter intervals accordingly. The posterior is then formed as a Gaussian mixture of the fits for each δ, with weights from the Bayesian information criterion. In the likelihood for each lag, the error scale is estimated together with γ and r_0. Every candidate lag has the same number of fitted parameters, so these weights reflect only how well each lag fits the data. This procedure places 98.0% of the weight on $\delta = 6$ yr among the candidates and gives posterior medians $\gamma = 0.338$ [0.299, 0.376] $\mathrm{mm\,SLE\,yr^{-1}\,{}^{\circ}C^{-1}}$ and $r_0 = 1.389$ [1.376, 1.401] $\mathrm{mm\,SLE\,yr^{-1}}$ (Fig. S1). The fit does not use the discharge rate at the end of the record separately, and it reproduces that rate. Over 2008–2018 the modeled rate is 1.466 $\mathrm{mm\,SLE\,yr^{-1}}$, compared with an observed 1.475 $\mathrm{mm\,SLE\,yr^{-1}}$, a difference smaller than the uncertainty of the observed rate.

The fitted $\delta = 6$ yr is an effective lag for spatially averaged forcing, and we therefore do not attempt to attribute it to a single process. It lies within the range of timescales spanned by the ocean-to-ice pathway. These timescales include the exchange of water between the continental shelf and the fjords, the response of glacier fronts to undercutting, and the inland spread of retreat to the flux gates, the fixed cross-sections near each glacier front where discharge is measured (*Straneo and Heimbach*, 2013; *Wood et al.*, 2021; *Robel et al.*, 2018). Because T_{ocean} is averaged over Greenland's peripheral waters, δ also absorbs the time required for a warming anomaly to become representative of conditions at outlet glaciers distributed around the ice sheet. Eq. 1 is linear and reversible, so it does not represent retreat that persists after ocean forcing declines, nor the control of fjord and bed topography on retreat. Glaciers calving on shallow ridges or standing in cold, shallow water have retreated little, while glaciers in deep fjords reached by warm Atlantic Water have retreated substantially (*Wood et al.*, 2021).

We validate the model against two records withheld from calibration, the discharge time series of *Mankoff et al.* (2021) spanning 1986–2023 and the time series from the Ice Sheet Mass Balance Intercomparison Exercise (IMBIE-3) spanning 1972–2023 (*Otosaka et al.*, 2026).

The validation with *Mankoff et al.* (2021) is informative but weak, because the *Mankoff et al.* (2021) time series shares its underlying raw data with the *Mouginot et al.* (2019) calibration and differs from it only in processing. IMBIE-3 combines three kinds of estimates. Input-output estimates subtract discharge from SMB, and those in IMBIE-3 include *Mouginot et al.* (2019) and *Mankoff et al.* (2021). Altimetry estimates, from satellite measurements of ice-surface height since 1992, and gravimetry estimates, from satellite measurements of Earth's gravity field since 2002, are independent of the calibration. Before 1992, IMBIE-3 rests on input-output estimates alone. Its discharge series is total mass balance minus the mean SMB of three regional climate models, one of which is RACMO, the model that also supplies the *Mouginot et al.* (2019) SMB. The independent part of the IMBIE-3 test is therefore the total mass balance after 1992. Our model falls within the uncertainties of the validation data for discharge and total mass balance (Fig. 2). Because the SMB part of the model is fitted to the *Mouginot et al.* (2019) SMB (Section 2.2.1), the total mass balance mainly tests the discharge model. The SMB emulator alone is tested against observed SMB in the Supplementary Material. The validation results suggest that our simple model, with two fitted parameters and a selected lag, captures cumulative GrIS discharge during the observational record and its sensitivity to ocean temperatures in Greenland's peripheral waters.

To our knowledge, no published study reports a Greenland-wide sensitivity of discharge rate to ocean temperature, so we derive a simple dimensional scaling to contextualize the inferred value $\gamma = 0.338\ \mathrm{mm\,SLE\,yr^{-1}\,{}^{\circ}C^{-1}}$. For a single outlet glacier i, the submarine melt rate scales approximately linearly with ocean thermal forcing, the ocean temperature above the in situ freezing point. We treat a change in T_{ocean} as an equal change in thermal forcing, so that

$$\Delta \dot{m}_i = \kappa_i \, \Delta T_{\mathrm{ocean}}, \tag{3}$$

where κ_i is the melt sensitivity to thermal forcing averaged over the submerged calving face. Plume parameterizations calibrated with Greenland observations (*Xu et al.*, 2013; *Rignot et al.*, 2016; *Slater and Straneo*, 2022) give a maximum melt sensitivity of 90–260 $\mathrm{m\,yr^{-1}\,{}^{\circ}C^{-1}}$ near the grounding line, where the glacier loses contact with its bed. The face-averaged rate is two to three times smaller (*Wood et al.*, 2021). Dividing by 2.5, we take κ_i = 36–104 $\mathrm{m\,yr^{-1}\,{}^{\circ}C^{-1}}$, with a geometric mean of 61 $\mathrm{m\,yr^{-1}\,{}^{\circ}C^{-1}}$. Submarine melt thins and undercuts the terminus, which triggers calving and retreat of the grounding line. The resulting increase in ice flux is larger than the direct submarine melt flux by a factor μ. Neither flux is measured ice-sheet wide, so we take μ = 3–10 as an assumed range. For comparison, submarine melt beneath floating termini removes 5–85% of grounding-line discharge (*Enderlin and Howat*, 2013), and ice sheet projections that exclude ocean-induced undercutting may underestimate mass loss by at least a factor of 2 (*Wood et al.*, 2021). The flux perturbation from a single glacier with submerged calving-front area A_{f} is then

$$\Delta \dot{Q}_i = \mu_i \, \rho_{\mathrm{ice}} \, A_{\mathrm{f},i} \, \kappa_i \, \Delta T_{\mathrm{ocean}}, \tag{4}$$

where $\rho_{\mathrm{ice}} = 900\ \mathrm{kg\,m^{-3}}$ is the density of ice. Summing over Greenland's $N = 219$ marine-terminating outlets (*Mouginot et al.*, 2019), we approximate the scale of γ as

$$\widetilde{\gamma} \sim \frac{1}{A_{\mathrm{ocean}} \, \rho_{\mathrm{fw}} \, \Delta T_{\mathrm{ocean}}} \sum_{i=1}^{N} \Delta \dot{Q}_i, \tag{5}$$

where $A_{\mathrm{ocean}} = 3.625 \times 10^{14}$ m^2 is the ocean area and $\rho_{\mathrm{fw}} = 1000$ kg m^{-3} is the density of freshwater. Together, these convert added ice mass to SLE. Submerged calving-front areas of 0.9–2.7 km^2 are reported for five West Greenland outlets (*Rignot et al.*, 2016), and we take this range as representative of all 219 outlets. Evaluating Eq. 5 across the full parameter ranges gives

$$\widetilde{\gamma} = 0.05\text{–}1.5 \text{ mm SLE yr}^{-1}\,^{\circ}\text{C}^{-1}, \tag{6}$$

which contains our calibrated $\gamma = 0.338$ mm SLE yr^{-1} °C^{-1}. We report Eq. 6 only as an order-of-magnitude check on the inferred value, to build intuition. The range spans a factor of 29 because it multiplies three uncertain quantities, μ, A_{f}, and κ, none of which is measured across the whole ice sheet. The estimate also treats these quantities as uncorrelated across outlets. In fact, the deepest and widest fronts sit in the warmest fjords, which would push the estimate upward. Conversely, two of the five glaciers with measured front areas are among the fastest in Greenland (*Rignot et al.*, 2016), so the adopted areas likely exceed the mean over all 219 outlets, most of which are small. Finally, surface meltwater that drains beneath glaciers strengthens the plumes that melt their fronts, and this runoff rises along with ocean temperature. The calibrated γ is therefore an effective sensitivity that also captures part of the atmospheric influence on discharge. At the ice-sheet scale, atmospheric variability in submarine melting accounts for about 30% of modeled retreat and dynamic sea-level contribution (*Slater and Straneo*, 2022).

2.1.1.2 Surface mass balance

For Greenland SMB, the forecast uses a statistical emulator of the regional climate model MARv3.12, driven by seven of the CMIP6 global climate models downscaled with MAR by *Boberg et al.* (2026). For each global model, we fit the annual SMB contribution as a quadratic in that model's GMST anomaly T relative to 1995–2005, smoothed with an 11-yr centered mean, with AR(1) residuals, $\dot{H}_{\mathrm{smb}} = a_{\mathrm{gris}}T^2 + b_{\mathrm{gris}}T + c_{\mathrm{gris}}$, pooling its historical and SSP simulations (Supplementary Material, Figs. S6–S8). Each Monte Carlo member draws one global model, with equal weight, and one posterior draw of a_{gris} and b_{gris}. Driven by observed GMST, the emulator does not reproduce a slowly varying departure of the observed SMB from its response to warming. Relative to 1995–2005, the *Mouginot et al.* (2019) SMB lies about 81 Gt yr^{-1} above the emulator over 1972–1985 and 106 Gt yr^{-1} below it over 2010–2018 (Supplementary Material). We represent this departure as a drift state $s(t)$ that relaxes toward zero over a timescale τ_{s}, so that $\dot{H}_{\mathrm{smb}} = a_{\mathrm{gris}}T^2 + b_{\mathrm{gris}}T + c_{\mathrm{gris}} + s(t) + \varepsilon(t)$, where ε is year-to-year weather. For each member, c_{gris} and one path of s are drawn from their Kalman-filter posterior given the *Mouginot et al.* (2019) SMB over 1972–2018. The 47-year record does not constrain τ_{s}. We therefore set it with a lognormal prior whose 90% range is 5 to 25 yr, which allows about two relaxation times within the record, and obtain $\tau_{\mathrm{s}} = 12$ [6, 26] yr. In 2018 the observed SMB minus the emulator is -24 [-163, 113] Gt yr^{-1}. After 2018 the state continues as a first-order autoregressive process whose mean relaxes toward zero. The drift state does not change the emulator's response to warming. We also include the SMB–elevation feedback, which adds $8 \pm 5\%$ (1σ, with negative draws set to zero) to SMB loss in the reference simulation of *Fettweis et al.* (2013). We scale this fraction by each member's cumulative SMB anomaly since 2000. Because MAR is the only regional climate model in the training set, the spread between regional climate models is not sampled (*Glaude*

et al., 2024). When each global model is left out in turn, the other six predict its 2015–2100 SMB contribution within the 90% predictive range for all 13 withheld simulations.

2.1.2 Thermosteric

The ocean is the dominant reservoir for the excess heat accumulated by the climate system (*Cheng et al.*, 2017), and the resulting thermal expansion of seawater, the thermosteric contribution, is currently the largest single contributor to observed sea-level rise (*Frederikse et al.*, 2020). A simple model of thermal expansion must represent both heat exchange between the ocean and the atmosphere and the transport of heat within the ocean. The ocean warms with a delay relative to the atmosphere. Heat enters at the sea surface and is carried downward by turbulent mixing and by the overturning circulation, so the temperature of the interior reflects the history of surface warming. Downward transfer spans a wide range of timescales. The upper few hundred meters exchange heat with the atmosphere over years to decades, whereas water below the thermocline is renewed over centuries. We therefore represent the ocean as two layers, a fast upper layer coupled to the atmosphere and a slow deep layer coupled only to the upper layer, following the structure of *Geoffroy et al.* (2013). This structure reproduces both the lag between surface warming and expansion and the continued expansion that follows a stabilization of surface temperature, because the deep layer continues to draw heat from the upper layer after the upper layer has equilibrated.

The model combines elements of earlier simple models. Its structure follows two-layer energy-balance models of the climate system (*Gregory*, 2000; *Held et al.*, 2010; *Geoffroy et al.*, 2013), in which a fast upper layer exchanges heat with the atmosphere and a slow deep layer exchanges heat with the upper layer. Those models are driven by radiative forcing. Ours is driven by GMST, as are semi-empirical models in which sea level relaxes toward an equilibrium set by temperature (*Grinsted et al.*, 2010; *Mengel et al.*, 2016; *Wong et al.*, 2017). Radiative forcing enters our model only through a term for volcanic eruptions, because GMST does not capture their effect on the ocean. In those models the thermosteric contribution has a single relaxation time, 82 to 1290 yr in *Mengel et al.* (2016), and an equilibrium sensitivity taken from long-term model integrations. In our model the delay acts on the thermal state of each layer rather than on sea level, and sea level follows from the thermal state through the equation of state of seawater, as in the layered thermal expansion of the MAGICC sea-level model (*Nauels et al.*, 2017). Models driven by ocean heat content instead convert heat to sea level with an expansion efficiency that is fixed in time (*Kuhlbrodt and Gregory*, 2012; *Fox-Kemper et al.*, 2021). Multi-layer energy-balance models driven by radiative forcing have been calibrated against observed ocean heat content in separate depth ranges (*Skeie et al.*, 2014; *Nadiga and Urban*, 2019). Our model applies this depth-resolved calibration to a model driven by GMST, in which each layer is compared directly with the observed thermosteric record for its own depth range, the upper layer with 0–700 m and the deep layer with 700–2000 m.

Consider first an upper layer of thickness $h_{\mathrm{u}} = 700$ m warmed by $\Delta\theta(z,t)$. The resulting rise in the steric height of the ocean is

$$H_{\mathrm{u}} = \int_0^{h_{\mathrm{u}}} \int_0^{\Delta\theta} \alpha_{\mathrm{sw}}\, d\theta'\, dz, \tag{7}$$

where $\alpha_{\mathrm{sw}}(z)$ is the thermal expansion coefficient of seawater, which varies with depth because it depends on temperature and pressure. The temperature anomaly profile is defined as $\Delta\theta = S_{\mathrm{u}}\varphi$, where $S_{\mathrm{u}}(t)$ is the amplitude of the anomaly and $\varphi(z)$ is a dimensionless shape factor. Expanding $\alpha_{\mathrm{sw}} = \alpha_{0,\mathrm{sw}}(z) + \alpha'_{\mathrm{sw}}\theta'$, with $\alpha'_{\mathrm{sw}} = \partial\alpha_{\mathrm{sw}}/\partial\theta$, we rewrite Eq. 7 as

$$H_{\mathrm{u}} = a_{\mathrm{u}} S_{\mathrm{u}}^2 + b_{\mathrm{u}} S_{\mathrm{u}}, \tag{8}$$

where $a_{\mathrm{u}} = \frac{1}{2}\int_0^{h_{\mathrm{u}}} f\alpha'_{\mathrm{sw}}\varphi^2 dz$ and $b_{\mathrm{u}} = \int_0^{h_{\mathrm{u}}} f\alpha_{0,\mathrm{sw}}\varphi dz$, and $f(z)$ is the ocean area at depth z as a fraction of the ocean surface area. The coefficient b_{u} sets the expansion of the upper layer per degree of warming, and a_{u} sets how that expansion grows with warming, because seawater expands more per degree when it is warmer. We do not fit a_{u}. For an anomaly profile of fixed shape, $\varphi = A\,g(z)$ with a constant A, the two integrals give $a_{\mathrm{u}} = \kappa_{\mathrm{sw}} b_{\mathrm{u}}^2$, where $\kappa_{\mathrm{sw}} = \frac{1}{2}\int_0^{h_{\mathrm{u}}} f\alpha'_{\mathrm{sw}} g^2 dz \big/ \left(\int_0^{h_{\mathrm{u}}} f\alpha_{0,\mathrm{sw}}\, g\, dz\right)^2$ depends only on the properties of seawater and the shape of the ocean basins. We compute $\kappa_{\mathrm{sw}} = 0.265\ \mathrm{m}^{-1}$ from the TEOS-10 equation of state (*IOC, SCOR and IAPSO*, 2010) and the 1995–2005 mean temperature and salinity of EN4 (*Good et al.*, 2013), for a profile that is uniform over 0–700 m, so that a_{u} follows b_{u} in each posterior sample. As GMST, $T(t)$, changes, the amplitude evolves as

$$\frac{dS_{\mathrm{u}}}{dt} = \frac{T - S_{\mathrm{u}}}{\tau_{\mathrm{u}}} + k_{\mathrm{v}} F_{\mathrm{v}}, \tag{9}$$

where τ_{u} is the characteristic time over which the thermal state S_{u} relaxes toward GMST, and the second term represents volcanic eruptions, as described below. We scale φ so that $S_{\mathrm{u}} = T$ at equilibrium, which places the ratio of upper-ocean warming to GMST in b_{u}. Both S_{u} and T are anomalies from the 1995–2005 mean. Without the volcanic term, Eq. 9 shows that S_{u} is an average of past temperatures, weighted toward the recent past over a time window set by τ_{u}. The upper layer therefore smooths year-to-year fluctuations in GMST and follows longer trends with a delay comparable to τ_{u}.

GMST alone does not capture the response of the upper ocean to volcanic eruptions. After a large eruption, stratospheric aerosol reduces the sunlight absorbed at the sea surface. Ocean heat content and sea level fall within about a year and recover over a decade or more (*Church et al.*, 2005), and part of the cooling reaches deeper layers, where it persists for decades (*Gleckler et al.*, 2006). GMST recovers within a few years, so a model driven by GMST alone misses most of this heat deficit. The second term in Eq. 9 adds it, with $F_{\mathrm{v}}(t)$ the volcanic effective radiative forcing of *Forster et al.* (2025). This forcing is an anomaly relative to the mean over 500 BCE to 1749 (*Forster et al.*, 2021), and its mean over 1850–2024 is close to zero. Energy conservation sets the coefficient, $k_{\mathrm{v}} = m_{\mathrm{v}} c_{\mathrm{v}}/b_{\mathrm{u}}$. Here $c_{\mathrm{v}} = 1.57$ mm per $\mathrm{W\,m^{-2}\,yr}$ is the 0–700 m thermosteric rise produced when a forcing of $1\ \mathrm{W\,m^{-2}}$ over the ocean, 71% of the Earth's surface, is sustained for one year. It uses the expansion efficiency of the NOAA 0–700 m record, $0.137\ \mathrm{m\,YJ^{-1}}$ ($1\ \mathrm{YJ} = 10^{24}$ J), the ratio of the thermosteric trend to the heat content trend over 1955–2010 (*Levitus et al.*, 2012). Dividing by b_{u} converts sea level to the thermal state S_{u}. The multiplier m_{v} has the prior $\mathcal{N}(1, 0.19^2)$, where $\mathcal{N}$ denotes a normal distribution, here with mean 1 and standard deviation 0.19. This prior combines the ±25% uncertainty (5–95%) of the volcanic forcing (*Forster et al.*, 2021) with the 12% uncertainty (one standard deviation) of the AR6 expansion efficiency (*Fox-Kemper et al.*, 2021). The forcing is zero after 2024, as in the AR6 scenarios, so the term affects the

projections only through the thermal state reached by 2024. For 2022–2024 it includes a small net positive forcing from the eruption of Hunga Tonga–Hunga Ha'apai, because the water vapor injected into the stratosphere outweighs the aerosol, although *Forster et al.* (2025) note that other studies find this net forcing to be negative or close to zero.

The deep layer follows the same construction, with $\Delta\theta_{\mathrm{d}}(z,t) = S_{\mathrm{d}}(t)\,\varphi_{\mathrm{d}}(z)$ over 700–2000 m depth and $b_{\mathrm{d}} = \int_{h_{\mathrm{u}}}^{h_{\mathrm{d}}} \alpha_{0,\mathrm{sw}}\,\varphi_{\mathrm{d}}\,dz$, with $h_{\mathrm{d}} = 2000$ m. We drop the quadratic term because S_{d} lags well behind S_{u} ($\tau_{\mathrm{d}} \gg \tau_{\mathrm{u}}$). At the posterior median τ_{u}, S_{d} rises about 1.0 °C above its 1995–2005 level by 2100 in the 3.6 °C scenario, compared with about 3.0 °C for S_{u}, and the 20-year 700–2000 m record cannot constrain a second quadratic coefficient. Warming from GMST reaches the deep layer only after passing through the upper layer, so the thermal state S_{d} relaxes toward S_{u} as

$$\frac{dS_{\mathrm{d}}}{dt} = \frac{S_{\mathrm{u}} - S_{\mathrm{d}}}{\tau_{\mathrm{d}}}, \tag{10}$$

where τ_{d} has a lognormal prior fitted to the ratio of deep-ocean heat capacity to the heat-exchange coefficient in the 16 CMIP5 models of *Geoffroy et al.* (2013, Table 4), with a median of 133 yr and a 90% range of 62 to 288 yr, because the 20-year 700–2000 m record is too short to constrain a centennial relaxation time. Adding this layer gives the total steric response for the upper 2000 m as

$$H_{\mathrm{ud}}(t) = H_{\mathrm{u}}(t) + b_{\mathrm{d}}\,S_{\mathrm{d}}. \tag{11}$$

Finally, to account for expansion below 2000 m depth, we define

$$H_{\mathrm{full}}(t) = H_{\mathrm{ud}}(t) + \dot{H}_{\mathrm{b}}(t - t_{\mathrm{ref}}), \tag{12}$$

where $\dot{H}_{\mathrm{b}} = 0.12 \pm 0.04\,\mathrm{mm\,yr^{-1}}$ (one standard deviation, 1991–2010) (*Desbruyères et al.*, 2016) is a constant rate for depths below 2000 m, referenced to year $t_{\mathrm{ref}} = 2000$. This layer is not observed at annual resolution, so its rate is adopted rather than calibrated.

We force the model with the Berkeley Earth GMST record (*Rohde and Hausfather*, 2020) and the volcanic forcing, and fit it jointly to thermosteric sea-level estimates (*Levitus et al.*, 2012) at 0–700 m (1955–2024), compared with H_{u}, and at 700–2000 m (2005–2024), compared with $b_{\mathrm{d}}S_{\mathrm{d}}$. The 700–2000 m record is the difference between the 0–2000 m and 0–700 m records in the same years. We fit this difference rather than the 0–2000 m record because the 0–2000 m record contains the 0–700 m data, which would otherwise enter the fit twice over 2005–2024. We multiply both NOAA records by 1.14 to correct the low bias identified by *Li et al.* (2022). We use Bayesian inference (*Tarantola*, 2005) in which Markov chain Monte Carlo (MCMC) sampling draws parameter combinations $\boldsymbol{\phi} = (b_{\mathrm{u}}, b_{\mathrm{d}}, \tau_{\mathrm{u}}, \tau_{\mathrm{d}}, m_{\mathrm{v}}, \sigma_{\mathrm{extra,u}}, \sigma_{\mathrm{extra,d}}, \rho)$ with probability density proportional to the posterior (Fig. S2 of the Supplementary Material). For each record $j \in \{\mathrm{u}, \mathrm{d}\}$, $\mathbf{r}_j(\boldsymbol{\phi})$ is the vector of residuals after subtracting the record's mean, and the model's mean over the same years, which removes each record's arbitrary reference level. We assume that the residuals are normally distributed about the model with covariance

$$\mathbf{C}_j = \mathrm{diag}\big(\sigma_{j,i}^2\big) + \sigma_{\mathrm{extra},j}^2\,\mathbf{R}_j, \qquad (\mathbf{R}_j)_{ik} = \rho^{|t_i - t_k|}, \tag{13}$$

where $\sigma_{j,i}$ is the observational error of point i in year t_i, the quadrature sum of the formal measurement error and a structural uncertainty equal to 5% of the bias-corrected anomaly,

retained after the bias correction. These errors are independent between years. The noise term $\sigma_{\mathrm{extra},j}$ absorbs misfit beyond the reported observational error, separately for each record. It represents variability in ocean heat content that the model does not resolve and that persists from one year to the next, so we correlate it between years as a first-order autoregressive process with coefficient ρ, shared by both records. The relaxation time of the upper layer, τ_{u}, has an informative prior, $\ln \tau_{\mathrm{u}} \sim \mathcal{N}(\ln 8, 0.5^2)$ with τ_{u} in years, so that $\mu_{\tau_{\mathrm{u}}} = \ln 8$ and $s_{\tau_{\mathrm{u}}} = 0.5$. The prior median is 8 yr and its 90% interval is [3.5, 18] yr, consistent with the ratio of upper-layer heat capacity to the heat-exchange coefficient, which spans 4 to 17 yr across the CMIP5 models in *Geoffroy et al.* (2013, Table 4), with a multi-model mean of 10 yr. The prior on τ_{d} has $\mu_{\tau_{\mathrm{d}}} = \ln 133$ and $s_{\tau_{\mathrm{d}}} = 0.47$, and the prior on m_{v} has standard deviation $s_m = 0.19$. Adding the negative log-priors gives the objective function

$$\mathcal{J}(\boldsymbol{\phi}) = \sum_j \left[\mathbf{r}_j^{\mathsf{T}} \mathbf{C}_j^{-1} \mathbf{r}_j + \ln\det \mathbf{C}_j\right] + \left(\frac{\ln \tau_{\mathrm{u}} - \mu_{\tau_{\mathrm{u}}}}{s_{\tau_{\mathrm{u}}}}\right)^2 + \left(\frac{\ln \tau_{\mathrm{d}} - \mu_{\tau_{\mathrm{d}}}}{s_{\tau_{\mathrm{d}}}}\right)^2 + \left(\frac{m_{\mathrm{v}} - 1}{s_m}\right)^2, \quad (14)$$

which is twice the negative log-posterior up to an additive constant. The $\ln\det \mathbf{C}_j$ term remains because $\mathbf{C}_j$ depends on the sampled $\sigma_{\mathrm{extra},j}$ and ρ. The parameters b_{u} and b_{d} have flat priors constrained to be non-negative, $\ln \sigma_{\mathrm{extra,u}}$ has a flat prior, and ρ has a flat prior on $(-0.99, 0.99)$. The prior on $\sigma_{\mathrm{extra,d}}$ is flat in $\sigma_{\mathrm{extra,d}}$ itself, on (0, 10] mm. The 700–2000 m residuals lie mostly within the reported errors, so the likelihood remains finite as $\sigma_{\mathrm{extra,d}}$ approaches zero, where a prior flat in its logarithm would be improper. The MCMC sampling uses the emcee package with 64 walkers, 6000 burn-in steps, and 30,000 production steps. The longest integrated autocorrelation time is 215 steps, about 1/140 of the chain length, giving at least 8900 effective samples per parameter. The parameters b_{u} and τ_{u} are partly degenerate in the likelihood, because a slower upper-layer response can be offset by a larger sensitivity to fit the same trajectory. This degeneracy is broken by τ_{u}'s informative prior. The posterior medians and 90% credible intervals are $b_{\mathrm{u}} = 42.3$ [36.9, 51.1] mm °C^{-1}, which gives $a_{\mathrm{u}} = 0.47$ [0.36, 0.69] mm °C^{-2}, $b_{\mathrm{d}} = 70$ [38, 139] mm °C^{-1}, $\tau_{\mathrm{u}} = 12.2$ [7.2, 21.9] yr, $\tau_{\mathrm{d}} = 157$ [71, 345] yr, $m_{\mathrm{v}} = 1.05$ [0.79, 1.30], $\sigma_{\mathrm{extra,u}} = 1.81$ [1.43, 2.51] mm, $\sigma_{\mathrm{extra,d}} = 0.45$ [0.11, 0.87] mm, and $\rho = 0.60$ [0.37, 0.80] (Fig. S2). The posterior of m_{v} is only slightly narrower than its prior, so the size of the volcanic term is set mainly by energy conservation rather than by the fit. The calibrated model reproduces both records with RMS residuals of 1.99 mm for depths 0–700 m and 0.61 mm for 700–2000 m, against 40 and 7.2 mm of observed steric rise over the respective records (Fig. 3).

We validate the full-depth model in Eq. 12 against two global-mean steric sea-level records, both withheld from calibration. *Frederikse et al.* (2020) reconstruct steric sea level for 1900–2018. Over 1957–2018 they draw, for each ensemble member, from three gridded products built on the same archive of in situ temperature and salinity profiles (*Levitus et al.*, 2012; *Ishii et al.*, 2017; *Cheng and Zhu*, 2016). The first of these is the data our model is calibrated on, so over this period the comparison measures consistency between separate analyses of largely shared data. Before 1957, in situ coverage is too sparse to resolve global steric change, and *Frederikse et al.* (2020) instead use the *Zanna et al.* (2019) reconstruction of ocean heat storage. That reconstruction is inferred from sea-surface temperature anomalies and heat-transport pathways in an ocean reanalysis. *Zanna et al.* (2019) also supplies

their expansion below 2000 m over the full 1900–2018 period, whereas our model uses the constant rate of *Desbruyères et al.* (2016) for this layer. The pre-1957 segment is therefore the part of this comparison that is independent of the data used in calibration. Our model agrees with the reconstruction to an RMS difference of 2.97 mm over 1957–2018 and 5.00 mm before 1957 (Fig. 3c,f). Our second validation comes from *Dangendorf et al.* (2024) who assimilate 516 tide-gauge records and satellite altimetry in an ensemble Kalman smoother that solves for GMSL over 1900–2021. Their global-mean steric estimate is what remains of GMSL after prescribed barystatic (from *Frederikse et al.*, 2020), glacial isostatic adjustment, and inverse-barometer contributions are removed. No temperature or salinity measurement enters this calculation, so the comparison tests our model against a steric estimate that shares none of the in situ ocean data used to calibrate it. Our model agrees with their reconstruction to an RMS difference of 3.65 mm over 1957–2021 and 11.44 mm before 1957.

2.1.3 Glaciers

Projected 21st-century glacier mass loss scales approximately linearly with global warming (*Rounce et al.*, 2023; *Edwards et al.*, 2021). We therefore model the rate of the glacier contribution to sea level as linear in GMST,

$$\dot{H}_{\mathrm{gl}} = b_{\mathrm{gl}} T + c_{\mathrm{gl}}, \tag{15}$$

where T is the calendar-year mean Berkeley Earth GMST anomaly relative to 1995–2005 (*Rohde and Hausfather*, 2020), b_{gl} is the sensitivity of the loss rate to warming, and c_{gl} is the loss rate at the 1995–2005 mean GMST. To treat uncertainty the same way as in the other components, we calibrate the model ourselves by Bayesian inference against the annual global glacier mass changes of the Glacier Mass Balance Intercomparison Exercise (GlaMBIE) for 2000–2023 (*The GlaMBIE Team et al.*, 2025). The GlaMBIE global total sums all 19 regions of the Randolph Glacier Inventory, including the peripheral glaciers of Greenland and Antarctica. Because the Greenland SMB emulator covers only the contiguous ice sheet (Section 2.2.1), the Greenland peripheral glaciers enter the projections only through this component. GlaMBIE's global total also includes the Antarctic peripheral glaciers (0.05 mm yr^{-1} over 2000–2023), part of which is also in the IMBIE-3 gravimetry estimates for Antarctica, so the two may overlap by up to that amount. We fit the time-integrated form of Eq. 15, which gives the cumulative contribution,

$$H_{\mathrm{gl}}(t) = b_{\mathrm{gl}} \int_{t_0}^{t} T(t')\,\mathrm{d}t' + c_{\mathrm{gl}}(t - t_0) + H_{\mathrm{gl},0} + \varepsilon_{\mathrm{gl}}(t), \tag{16}$$

where $t_0 = 2000$ is the baseline year and $H_{\mathrm{gl},0}$ is the modeled contribution in that year. The observational error $\varepsilon_{\mathrm{gl}}$ is normally distributed with covariance Σ_{gl}. Summing annual mass changes into a cumulative record correlates its errors in time, and Σ_{gl} is the covariance of the cumulative record implied by independent annual errors. GlaMBIE reports its uncertainties as 95% intervals, which we convert to standard deviations. The residuals scatter more widely than these uncertainties allow, with a reduced chi-square of 4.1, so we add a model-inadequacy term $\sigma_{\mathrm{gl,extra}}$ in quadrature to each annual uncertainty and fit it with the other parameters.

We apply a weakly informative prior $b_{\mathrm{gl}} \sim \mathcal{N}(0.61, 2.0^2)$ mm yr^{-1} °C^{-1}, truncated to $b_{\mathrm{gl}} \geq 0$. The prior mean is drawn from *Rounce et al.* (2023), whose end-of-century rates of glacier sea-level rise increase from 0.70 to 2.23 mm yr^{-1} as warming above pre-industrial increases from 1.5 to 4 °C, a slope of 0.61 mm yr^{-1} °C^{-1}. The standard deviation is wide enough that the prior is only weakly informative, and widening it fiftyfold leaves the posterior unchanged. The non-negativity truncation recognizes that glaciers lose mass as the climate warms.

Left free, c_{gl} makes the fitted model gain glacier mass in most years before 1950 and in 18 of the 30 years from 1950 to 1979. In reality, glaciers entered the industrial era close to balance with the slowly cooling climate of the preceding centuries (*Roe et al.*, 2021). We therefore require the loss rate to be zero at a balance temperature $T_{\mathrm{gl,eq}}$, which gives $c_{\mathrm{gl}} = -b_{\mathrm{gl}} T_{\mathrm{gl,eq}}$. The model we fit is

$$H_{\mathrm{gl}}(t) = b_{\mathrm{gl}} \int_{t_0}^{t} [T(t') - T_{\mathrm{gl,eq}}] \, \mathrm{d}t' + H_{\mathrm{gl},0} + \varepsilon_{\mathrm{gl}}(t). \tag{17}$$

The prior on $T_{\mathrm{gl,eq}}$ is normal, centered on the 1850–1879 mean GMST of −0.86 °C relative to 1995–2005, with a standard deviation of 0.15 °C. Equation 17 neglects the adjustment of glacier geometry to climate, which sets each glacier's response time (*Roe et al.*, 2021), so it represents the limit of long response times.

The MCMC sampling uses the emcee package with 32 walkers, 2000 burn-in steps, and 4000 production steps. The posterior medians are $b_{\mathrm{gl}} = 0.76$ [0.60, 0.99] mm yr^{-1} °C^{-1}, $T_{\mathrm{gl,eq}} = -0.78$ [−1.02, −0.56] °C, $c_{\mathrm{gl}} = 0.59$ [0.52, 0.67] mm yr^{-1}, and $\sigma_{\mathrm{gl,extra}} = 0.18$ [0.13, 0.26] mm yr^{-1} (Fig. 4a). With c_{gl} free, the fit gives $b_{\mathrm{gl}} = 0.99$ [0.57, 1.45] mm yr^{-1} °C^{-1} and $c_{\mathrm{gl}} = 0.53$ [0.41, 0.66] mm yr^{-1}. The 2000–2023 record spans only about 0.8 °C of GMST, so it constrains the present loss rate more tightly than the sensitivity. The balance constraint adds information from the pre-industrial climate, which lowers the median of b_{gl} by about a quarter and narrows its interval.

We validate the glacier model against the global glacier mass changes of *Zemp et al.* (2019) for 1962–1999, which are derived from glaciological and geodetic observations and are not used in the calibration. Over these 38 years the model gives a mean rate of 0.30 [0.17, 0.40] mm yr^{-1}, compared with an observed 0.32 [−0.02, 0.66] mm yr^{-1}, where the observed interval is the 90% range derived from the 95% uncertainties reported by *Zemp et al.* (2019) (Fig. 4a). Because the model with c_{gl} free predicts glacier mass gain in most years before the late 1970s, this comparison also tests the balance constraint. We compare only the years before 2000. Over 2000–2016, when both records exist, the loss rate of *Zemp et al.* (2019) exceeds that of GlaMBIE by 0.18 mm yr^{-1}, so a comparison there would measure the disagreement between the two records rather than the skill of the model.

2.1.4 Antarctic Peninsula

The Antarctic Peninsula component uses the same level-space Bayesian framework as glaciers, fit at annual resolution to the IMBIE-3 Peninsula time series spanning 1979–2023 (45 annual points; *Otosaka et al.*, 2026). We choose the linear rate–temperature model $\dot{H}_{\mathrm{p}} = b_{\mathrm{p}} T + c_{\mathrm{p}}$, which captures the combined response of SMB and ice discharge to warming. IMBIE-3 attributes the Peninsula mass loss over 1979–2023 to ice discharge (*Otosaka et al.*, 2026),

which increased after the Larsen A and B ice shelves collapsed in response to regional atmospheric warming (*Scambos et al.*, 2004). Here $H_{\mathrm{p},0}$ is the model level at the start of the record (1979), fitted with the other parameters, and the likelihood uses the covariance of this cumulative record implied by independent annual IMBIE-3 errors. As for glaciers, we also fit a model-inadequacy term $\sigma_{\mathrm{p,extra}}$ to account for annual scatter beyond the IMBIE-3 reported uncertainty, which lowers the reduced chi-square from 2.63 to 1.00. The MCMC sampling uses the emcee package with 32 walkers, 2000 burn-in steps, and 4000 production steps. The priors are $b_{\mathrm{p}} \sim \mathcal{N}(0,\, 2.0^2)$ mm yr^{-1} °C^{-1} and $c_{\mathrm{p}} \sim \mathcal{N}(0.3,\, 1^2)$ mm yr^{-1}. The posterior median sensitivity is $b_{\mathrm{p}} = 0.045$ [−0.022, 0.107] mm yr^{-1} °C^{-1}, and the rate at the 1995–2005 mean GMST is $c_{\mathrm{p}} = 0.056$ [0.038, 0.072] mm yr^{-1} (Fig. 4b). The 90% credible interval for b_{p} spans zero, so these data do not, on their own, resolve the sign of the temperature sensitivity.

2.1.5 East Antarctica

We fit the linear model $\dot{H}_{\mathrm{e}} = b_{\mathrm{e}}T + c_{\mathrm{e}}$ to the IMBIE-3 East Antarctic Ice Sheet (EAIS) record (1979–2023, 45 annual points; *Otosaka et al.*, 2026) in the same level-space framework as the Peninsula, with a weak prior $b_{\mathrm{e}} \sim \mathcal{N}(-0.17,\, 0.5^2)$ mm yr^{-1} °C^{-1} and no model-inadequacy term (reduced chi-square 0.90). The posterior median sensitivity is $b_{\mathrm{e}} = -0.028$ [−0.160, 0.102] mm yr^{-1} °C^{-1} (Fig. 4c). This interval spans zero, so the record does not constrain the sensitivity. The forecast instead adopts the snowfall-driven mass gain of $C_T = 60 \pm 20$ Gt yr^{-1} per °C of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013), which converts at 362.5 Gt per mm to −0.17 mm yr^{-1} °C^{-1} of sea level and also sets the prior mean (Section 2.2.1).

2.1.6 West Antarctica

Unlike for glaciers, the Antarctic Peninsula, and EAIS, we do not fit a rate–temperature regression to WAIS over the observational record, because ocean heat has driven most WAIS mass loss and ocean observations are too sparse to support a fit to ocean temperature. We instead fit a rate that changes linearly in time to the 1979–2023 IMBIE-3 record (*Otosaka et al.*, 2026), so the cumulative contribution is quadratic in time. As for the Antarctic Peninsula, the likelihood is that of the annual increments of the cumulative record, each with the reported IMBIE-3 uncertainty plus a fitted excess scatter. The acceleration a_{w} has a zero-mean prior, $a_{\mathrm{w}} \sim \mathcal{N}(0,\, 0.05^2)$ mm yr^{-2}. The fit yields an acceleration of 0.0122 [0.0099, 0.0143] mm yr^{-2} and a rate in year 2000 of 0.247 [0.219, 0.273] mm yr^{-1} (Fig. 4d). *Davison et al.* (2023) report snowfall anomalies in the Amundsen Sea Embayment over 2009–2013 and 2019–2020 that modulate this record. Removing them lowers the extrapolated contribution at 2100 from 86 to 79 mm, well within the S1 uncertainty described below, so we fit the record as reported. We set the WAIS rate to zero before 1979, when the IMBIE-3 record begins.

2.2 Forecasting methods

This section describes how the forecast joins the observed satellite-era trend to the component models, how uncertainty is propagated, and how each component is linked to projected

GMST. The forecasts presented here use the IPCC AR6 GMST trajectories for different emission scenarios, though they can be applied to any GMST trajectory. We focus on the policy-relevant emission scenarios, labeled as in the Introduction by their AR6 central-estimate warming over 2081–2100: 1.8, 2.7, and 3.6 °C. We initialize the GMSL forecast using the GMSL level, rate, and acceleration observed in the satellite era and blend in the component-model sum in rate space, such that

$$\dot{H}_{\mathrm{forecast}} = w\dot{H}_{\mathrm{obs}} + \big[1 - w\big]\dot{H}_{\mathrm{comp}}, \tag{18}$$

where $\dot{H}_{\mathrm{obs}}$ is the rate of our quadratic fit to the NASA satellite altimetry record over 1993–2025, following *Nerem et al.* (2022) and *Hamlington et al.* (2024), and $\dot{H}_{\mathrm{comp}}$ is the rate of the summed component projections. The blending weight $w(t)$ declines smoothly from 1 to 0,

$$w = 1 - \Theta\left(\frac{t - t_{\mathrm{r}}}{\tau_{\mathrm{r}}}\right), \tag{19}$$

where $\Theta(x) = (1 + e^{-x})^{-1}$ is the standard logistic function, $t_{\mathrm{r}} = 2035$, and $\tau_{\mathrm{r}} = 5$ yr. At the forecast origin $w = 0.87$. At $t \ll t_{\mathrm{r}}$, $w \approx 1$ and the forecast follows the observed quadratic trajectory. At $t \gg t_{\mathrm{r}}$, $w \approx 0$ and the forecast follows the component models. The weight falls from 0.9 to 0.1 over $t_{\mathrm{r}} \pm 2.2\tau_{\mathrm{r}}$, a span of about 22 yr centered on 2035. We chose $t_{\mathrm{r}} = 2035$, 10 yr after the forecast origin, or about one third of the length of the 1993–2025 satellite altimetry record, a period when the extrapolation of observations should have strong forecasting skill. The tapering recognizes the expected reduction in forecasting skill of the simple data extrapolation over timescales longer than roughly half of the observational window. Moving t_{r} 5 yr later, to about half the length of the altimetry record and arguably the latest viable choice, moves the 2050 and 2100 medians by 5 mm or less at 1.8–3.6 °C, so the results are insensitive to t_{r} between 2035 and 2040.

The GMSL forecast is obtained by integrating Eq. 18 forward from the level of the same quadratic fit at the forecast origin ($t_{\mathrm{f}} = 2025$, the last year of NASA altimetry data at the time of writing), relative to its level in 2000, so that

$$H_{\mathrm{forecast}}(t) = H_{\mathrm{obs}}(t_{\mathrm{f}}) + \int_{t_{\mathrm{f}}}^{t} \dot{H}_{\mathrm{forecast}}(t')\, dt'. \tag{20}$$

Each member starts from the level of its own draw of the quadratic fit at t_{f}, so the starting level carries the 6.4 mm (1σ) uncertainty of the fit and stays consistent with that member's early rate. Taking the starting level from the fit, rather than from the altimetry record at a single time, keeps interannual variability out of the starting level. Blending rates rather than levels keeps the forecast continuous with the quadratic fit at the forecast origin, so the offset between the component-sum level and the observations does not enter the forecast. Each component has its own 2000-member ensemble of level-space projections $H_k(t)$ (for WAIS, a random subsample of a 10,000-member ensemble), with parameters drawn independently from that component's posterior or, for components using prescribed literature sensitivities rather than data calibration, from the corresponding parameter distributions. Member i of every GMST-driven component follows the same GMST path. This path is the AR6 median plus z_i times the growth since 2024 of the AR6 standard deviation, which we take as the

half-width to the 5th or 95th percentile, according to the sign of z_i, divided by 1.645. The value z_i is one standard-normal draw held for the whole century and used in every scenario. Through 2024, all members follow observed GMST. After 2024, the AR6 paths are placed on the Berkeley Earth basis by one offset shared by every component and scenario, the mean difference between the AR6 SSP2-4.5 median and Berkeley Earth GMST over 2015–2024. The total level is formed sample by sample, $H_{\mathrm{comp},i}(t) = \sum_k H_{k,i}(t)$ for $i = 1, \ldots, 2000$, so parameter draws are independent across components while the within-scenario warming path is shared. The component rate $\dot{H}_{\mathrm{comp}}$ used in Eq. 18 is obtained by finite-differencing this summed ensemble in time before blending with the observational quadratic rate. The blending is done member by member across the 2000-member ensemble. The quadratic coefficients are drawn jointly from their fitted covariance, independently of the components, so the correlations within the component sum, including the shared warming path, carry into the forecast.

2.2.1 Component-level GMST dependence

Four components need an extra step to link them to projected GMST. Greenland ice discharge depends on the temperature of the ocean around Greenland rather than on GMST. Greenland SMB is emulated from a regional climate model (Section 2.1.1.2). The EAIS sensitivity is not constrained by the observed record (Section 2.1.5), so we adopt a literature value. Terrestrial water storage is adopted from AR6.

To project the GrIS discharge model forward in time, we relate ocean temperature to Greenland regional surface temperature through a transfer function. We then use an Arctic amplification factor, the ratio of Greenland regional to global warming, drawn from a normal distribution, to calculate the Greenland regional surface temperature from projected GMST under different emission scenarios. The subsurface ocean does not follow the surface from one year to the next. About their linear trends over 1970–2020, Greenland regional surface temperature varies from year to year with a standard deviation of 0.80 °C, and EN4 ocean temperature at 200–500 m with a standard deviation of 0.19 °C. A regression of annual ocean temperature on annual surface temperature would therefore underestimate the response of the ocean to sustained warming. We instead relate ocean temperature to a running average of past surface temperature,

$$T_{\mathrm{ocean}}(t) = \xi\, \overline{T}^{\mathrm{Gr}}_{\mathrm{surface}}(t) + \chi + \varepsilon_{\mathrm{go}}(t), \qquad \frac{d\overline{T}^{\mathrm{Gr}}_{\mathrm{surface}}}{dt} = \frac{T^{\mathrm{Gr}}_{\mathrm{surface}} - \overline{T}^{\mathrm{Gr}}_{\mathrm{surface}}}{\tau_{\mathrm{go}}}, \tag{21}$$

where $T^{\mathrm{Gr}}_{\mathrm{surface}}$ is the Greenland regional surface temperature anomaly, the average over the Berkeley Earth land cells that lie in Greenland (*Rohde and Hausfather*, 2020), τ_{go} is the response time of the subsurface ocean, and $\varepsilon_{\mathrm{go}}$ is a zero-mean residual. Both temperatures are anomalies relative to the 1995–2005 baseline. We step the running average forward in annual increments from the start of the surface record in 1850. For each whole-year value of τ_{go} from 1 to 20 yr, we fit (ξ, χ) by generalized least squares with first-order autoregressive residuals over the 51 years of overlap, 1970 to 2020, and weight each value by its likelihood. The data rule out response times shorter than about 2 yr but constrain longer ones only weakly. The grid ends at 20 yr, where the likelihood has fallen to 1/22 of its maximum. The

fit gives $\tau_{\mathrm{go}} = 5$ [3, 16] yr and $\xi = 0.30$ [0.16, 0.55]. At the most likely response time, 4 yr, the residual has standard deviation $\sigma_{\varepsilon_{\mathrm{go}}} = 0.15\,^{\circ}\mathrm{C}$ and lag-1 autocorrelation 0.68. In the years the EN4 record does not cover, before 1970 and from 2021 to 2024, we apply Eq. 21 to the observed Greenland regional surface temperature. We carry the residual into those years and into the projections as first-order autoregressive noise.

For projections, the scenario GMST trajectories are converted to Greenland regional surface temperature using $\mathrm{AA} = 2.5 \pm 0.5$. The Arctic as a whole warmed nearly four times faster than the globe over 1979–2021 (*Rantanen et al.*, 2022), but the ratio of Greenland regional to global warming in the observations we use is lower and depends on the averaging window, spanning 2.0 to 2.7 for windows that start between 1970 and 1990 and end in 2024. Greenland regional surface temperatures are used in Eq. 21 to obtain projected T_{ocean}, which drives the lagged discharge model (Eqs. 1 and 2). Each of the 2000 Monte Carlo members draws $\mathrm{AA} \sim \mathcal{N}(2.5, 0.5^2)$ independently, τ_{go} from its likelihood weights and (ξ, χ) from their fitted covariance at that τ_{go}, and one autocorrelated path of $\varepsilon_{\mathrm{go}}$, together with one draw of (γ, r_0, δ) from the discharge posterior, so that Arctic amplification, transfer-function, and discharge parameter uncertainties all propagate into the projections. A single draw of $(\mathrm{AA}, \tau_{\mathrm{go}}, \xi, \chi, \gamma, r_0, \delta)$ is applied to all scenarios, so that differences between scenarios isolate the effect of the emissions pathway from parameter uncertainty.

For EAIS SMB, we use the linear model of Section 2.1.5 with the literature-derived accumulation sensitivity $C_T = 60 \pm 20$ Gt yr^{-1} °C^{-1} of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013), equivalent to $b_{\mathrm{e}} = -0.17$ mm yr^{-1} °C^{-1}. This sensitivity reflects the Clausius–Clapeyron scaling of snowfall with warming ($C_T > 0$ corresponds to EAIS mass gain through accumulation). Data from IMBIE-3 show EAIS in approximate balance (Fig. 4c), so we set $c_{\mathrm{e}} = 0$. The EAIS contribution is small (-13 to -24 mm at 2100 across the 1.8–3.6 °C scenarios) relative to the total, meaning the GMSL projection results of this study are largely insensitive to these parameter choices.

Terrestrial water storage (TWS) projections are adopted directly from IPCC AR6 rather than modeled here. Groundwater depletion, reservoir impoundment, and changes in land hydrology are outside the scope of this framework, and the data are too sparse for a data-driven model. At each IPCC time step we fit a normal distribution to the reported 5th, 17th, 50th, 83rd, and 95th percentiles. Each of the 2000 members takes a single quantile of these distributions for the whole century, and its path is linearly interpolated to annual resolution. For the hindcast and budget-closure diagnostics we use the *Frederikse et al.* (2020) TWS estimate over its full record, 1900–2018, with Monte Carlo paths whose errors accumulate from 2000, and hold it at its 2018 value thereafter. The observed GMSL records that these diagnostics test contain TWS, so the component sum must contain it too. Their TWS is derived from GRACE and GRACE-FO observations over 2003–2018, partitioned from the total land-mass signal. Before 2003 it is a composite of a reconstruction of climate-driven water storage and process models for reservoir impoundment and groundwater depletion rather than a measurement. Fig. S10 of the Supplementary Material shows the hindcast with TWS set to zero before 2003. TWS varies little among scenarios and its contribution is small relative to the total, so it does not materially affect the uncertainty budget.

2.2.2 West Antarctic Ice Sheet

West Antarctica dominates the uncertainty in sea-level projections because its major outlet glaciers, particularly Thwaites and Pine Island, lie close to a threshold at which small changes in ice geometry could trigger self-sustaining retreat (*Joughin et al.*, 2014; *Pattyn and Morlighem*, 2020; *Fricker et al.*, 2025). Whether that threshold has been, or will soon be, crossed depends on glacier-specific details that remain only partly observed. We do not attempt to constrain a GMST sensitivity for WAIS. Instead, we represent WAIS as a mixture of two scenarios. In Scenario 1 (S1), mass loss continues along the observed trend. In Scenario 2 (S2), loss accelerates well beyond it.

Given growing evidence that WAIS mass loss is primarily sensitive to internal ice dynamics and to deep-ocean temperatures whose 21st-century warming is largely committed regardless of emissions, we make our WAIS projections independent of the warming trajectory (*Naughten et al.*, 2023; *Coulon et al.*, 2024; *Rosier et al.*, 2025). This independence is limited to the 21st century, because warming over this century largely sets the Antarctic contribution in the centuries that follow (*Lowry et al.*, 2021; *Seroussi et al.*, 2024). Some studies find emissions dependence even before 2100, with major WAIS retreat beginning this century in some simulations under 3 °C of warming but not under 2 °C (*DeConto et al.*, 2021). An emulation of the ISMIP6 Antarctic ensemble extended to 2300, calibrated to the observed 2002–2021 Antarctic mass-loss rate, finds that higher emissions raise the 2100 Antarctic contribution with probability 0.89, but its posterior median for the whole ice sheet rises by only 5 mm from the 1.4 to the 2.7 °C scenario (*Lin et al.*, 2026). That shift is small compared with our S1 90% credible interval of 86 mm [−10, 182], and part of it arises from the SMB response that we carry separately for EAIS.

In S1, mass loss continues along the observed trajectory (Fig. 4d). Extrapolating the trend gives a median GMSL contribution of 86 mm by 2100, 1 mm above the ISMIP6 emulator median of 85 mm (*Edwards et al.*, 2021). A 45-year observational record cannot itself constrain how much additional error should accumulate from extrapolating a further 76 years beyond it, so a credible interval built from the fitted parameters alone would understate the uncertainty in this extrapolation. We instead treat the growth in uncertainty that the ISMIP6 emulator reports beyond the last observation as an external estimate of the extrapolation error. We add it to our parameter uncertainty as a prediction error (*Tarantola*, 2005), which widens our 90% credible interval at 2100 to [−10, 182] mm, closely matching the ISMIP6 emulator range of [−11, 182] mm.

In S2, mass loss accelerates well beyond currently observed rates. We define the S2 distribution at 2100 by its 5th percentile of 85 mm, equal to the S1 median, and its 95th percentile of 1.3 m, approximately the IPCC AR6 low-confidence Antarctic Ice Sheet 95th percentile at 2100 under the 4.4 °C scenario (*Fox-Kemper et al.*, 2021). The distribution is right-skewed. Its logarithm follows a skew-normal distribution with shape parameter $\lambda = 3$, reflecting the expectation that instability skews outcomes toward greater loss (*Robel et al.*, 2019). Varying λ from 0 to 6 moves the 2100 median of the mixture with $p(\mathrm{S1}) = 10\%$ from 0.30 to 0.20 m and its 95th percentile by 0.01 m. The trajectory of WAIS mass loss through time is set by a power law in time that starts near the observed rate and accelerates toward the 2100 value, with median exponent $\beta = 2$, for which the S2 95th percentile stays within the AR6 medium-confidence range for the Antarctic Ice Sheet until about 2050 and exceeds

it afterward.

Emerging research on physical processes not yet represented in ice-sheet models supports the possibility of loss well above the observed trajectory. Examples include seawater intrusion beneath grounded ice (*Rignot et al.*, 2024; *Bradley and Hewitt*, 2024) and the evolution of subglacial water pressure (*Zhao et al.*, 2025). The ISMIP6 ensemble itself implies far less loss. *Lin et al.* (2026) identify a set of ice-sheet model choices that remain consistent with the observed mass-loss rate and yield 157 mm [98, 254] for the whole ice sheet at 2100 under the 4.4 °C scenario. The 95th percentile of that range is roughly a fifth of our S2 95th percentile for WAIS alone. That ensemble largely omits marine ice cliff instability, subglacial hydrology, and ice damage, so the difference reflects processes absent from current ice-sheet models. The S2 95th percentile adopted here should therefore not be read as an upper bound on plausible loss, and Fig. 6 shows how the WAIS contribution changes with it.

The final probability density is a simple mixture of S1 and S2, weighted by the probability of S1, $p(\mathrm{S1})$ (Fig. 5). Its median, 95th percentile, and other quantiles are determined primarily by $p(\mathrm{S1})$ and the S2 95th percentile (Fig. 6). We hold the S2 95th percentile fixed and use $p(\mathrm{S1})$ as the control variable. We present two storylines at the limits of the IPCC terms "very likely" (90%) and "very unlikely" (10%): a slow-WAIS storyline with $p(\mathrm{S1}) = 90\%$, in which continued mass loss along the observed trajectory is very likely, and a fast-WAIS storyline with $p(\mathrm{S1}) = 10\%$, in which it is very unlikely.

3 Results

We first test the framework against observations and then present projections. Hindcasting skill matters for judging a model's projections, because it tests forecasting skill against data and indicates the horizon over which the projections can be trusted. We test the summed components against satellite altimetry (budget closure) and against GMSL reconstructions back to 1900 (hindcasts), then present projections for each component and for total GMSL.

3.1 Validation

We validate the model in two ways. First, we test whether the sum of the separately calibrated components accounts for the observed rate of GMSL rise (budget closure). Second, we run the model over the past century with observed GMST (a hindcast) and compare it with independent GMSL records.

3.1.1 Total GMSL budget closure

Satellite altimetry over 1993–2025 provides observations of GMSL (*Beckley et al.*, 2017; *Hamlington et al.*, 2020) that can be compared with the sum of the components (Fig. 7). The thermosteric, glacier, and Peninsula contributions come from the calibrated models driven by observed GMST, and Greenland discharge from its model driven by observed ocean temperature. WAIS and EAIS enter as the IMBIE-3 records (*Otosaka et al.*, 2026) through 2023, continued by the WAIS fit and the EAIS SMB model. Total GMSL is withheld

from calibration to serve as a validation data set. For Greenland SMB we use the RACMO-derived SMB of *Mouginot et al.* (2019) through 2018 and the emulator thereafter. Terrestrial water storage is adopted from *Frederikse et al.* (2020) and held at its 2018 value after their record ends. The uncharted-glacier term, described in the hindcast subsection below, is zero from 2000 onward.

The component sum accounts for 89–104% of the observed rate of GMSL rise over three epochs (1993–2025, 2000–2025, 2010–2025), indicating that the individually calibrated components together account for observed GMSL during the satellite altimetry record (Fig. 7). Because each component is calibrated independently, we treat their rate uncertainties as independent and combine them in quadrature. Each component's individual uncertainty comes directly from its Monte Carlo ensemble. For a given epoch, we fit a linear trend to every ensemble member and take the standard deviation of the resulting trends. Repeating this for each epoch lets the combined uncertainty differ from one epoch to the next, rather than using one fixed value throughout the satellite record. We add the uncertainty of the altimetry trend over the same epoch in quadrature, combining the published altimetry error covariance, the serial correlation of the residuals, and the uncertainty of the glacial isostatic adjustment correction. Errors on the cumulative records that enter the closure directly, IMBIE-3 for WAIS and EAIS, the observed Greenland SMB, and terrestrial water storage, are drawn as correlated annual increments. By this measure, the residual between the observed and summed rates is 0.44σ, 0.34σ, and 1.01σ of the combined uncertainty for 1993–2025, 2000–2025, and 2010–2025. The largest shortfall is over 2010–2025. It arises after 2018: the component sum accounts for about 102% of the observed rate over 2010–2018 and about 77% over 2018–2025, when the closure uses the SMB emulator in place of the observed Greenland SMB and holds terrestrial water storage at its 2018 value.

3.1.2 Hindcasting the level, rate, and transient sensitivity

We validate the combined GMST-sensitive model by two hindcasting approaches. In both, the GMST-dependent components are forced with observed GMST back to 1900 (*Rohde and Hausfather*, 2020), with an emphasis on post-1950 when increasingly reliable GMSL data are available. WAIS follows the IMBIE-3 record, and its rate is set to zero through 1979. Terrestrial water storage follows *Frederikse et al.* (2020) over 1900–2018. The glacier hindcast also includes glaciers that are missing from the inventories on which the GlaMBIE record is built, because they are too small to map or because they melted away after 1901. *Parkes and Marzeion* (2018) estimate that these uncharted glaciers contributed 0.17 to 0.53 mm yr^{-1} over 1901–1990, most of it early in the century, and 0.08 to 0.21 mm yr^{-1} over 1993–2010. We add this contribution as a prescribed series, sampled between their lower and upper bounds. Within each bound the rate declines linearly over 1901–1990, with a mean equal to their 1901–1990 value, and then holds their 1993–2010 rate. It does not depend on GMST and is set to zero from 2000, where GlaMBIE calibrates the glacier model.

The first comparison is between the GMSL level and rate from SPECTRUM and those of the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions, the IPCC AR6 compilation (*Fox-Kemper et al.*, 2021), which includes *Frederikse et al.* (2020) and an earlier version of the *Dangendorf et al.* (2024) reconstruction, and satellite altimetry (*Beckley et al.*, 2017). For the second comparison, we calculate the transient sensitivity of the SPECTRUM

output, the change in the rate of GMSL rise per degree of warming, $\alpha = d\dot{H}/dT$, where $\dot{H}$ is the rate of GMSL rise. For the level and rate comparison, we use the same rate-space blending scheme as the forecast (Section 2.2), reflected about the forecast origin so that the satellite-era quadratic dominates at late times and the component models at early times. The midpoint of the blend falls at 2015, the mirror of 2035 about $t_{\mathrm{f}} = 2025$, with $\tau_{\mathrm{r}} = 5$ yr unchanged. The observational term thus carries the same weight at 2025 as in the forecast and falls to 0.05 by 2000, so the hindcast over the reconstruction era tests the component models rather than restating the altimetry extrapolation.

Extending the hindcast back to 1900 shows distinct regimes relative to the various GMSL time series (Fig. 8a). Over the satellite era we compare rates over the window that each record covers, using the component sum without blending, because over this period the blend includes the satellite-era quadratic. The hindcast rate is 3.31 mm yr^{-1} compared with 3.35 mm yr^{-1} for the *Frederikse et al.* (2020) reconstruction (1993–2018), 3.36 compared with 3.17 mm yr^{-1} for the *Dangendorf et al.* (2024) reconstruction (1993–2021), 3.35 compared with 3.23 mm yr^{-1} for the IPCC AR6 observational compilation (1993–2020; *Fox-Kemper et al.* 2021), and 3.39 compared with 3.29 mm yr^{-1} for satellite altimetry (1993–2025; *Beckley et al.* 2017). Over 1950–2000 the hindcast rate is 0.75 mm yr^{-1} compared with 1.02, 1.01, and 1.28 mm yr^{-1} for the same three records, and the hindcast level at 1950 lies 11 to 23 mm above them. Of the hindcast rate, 0.21 mm yr^{-1} comes from the prescribed uncharted-glacier term. The hindcast includes the terrestrial water storage of *Frederikse et al.* (2020), which lowered GMSL by 18 mm from 1950 to 2000. Before 2003 this estimate combines a reservoir inventory with hydrological and groundwater models rather than observations. With terrestrial water storage set to zero before 2003, the 1950–2000 rate of the summed components would be 1.18 mm yr^{-1} (Fig. S10 of the Supplementary Material). In the budget of *Frederikse et al.* (2020), Greenland raised GMSL from 1950 to 2000 by 8 mm more than in our hindcast, of which about 3 mm is the Greenland peripheral glaciers that *Frederikse et al.* (2020) count with Greenland and we count with glaciers. Before 1972, their Greenland contribution comes only from *Kjeldsen et al.* (2015), which resolves a single mean rate of mass loss over 1900–1983. Before 1950, where only the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions extend, the divergence grows: the hindcast rate is 1.12 mm yr^{-1} compared with 1.39 and 1.64 mm yr^{-1} over 1900–1950, and the hindcast level at 1900 is 32–38 mm above the reconstructions, relative to the 2000 baseline. Before 1950, the median line from SPECTRUM stays within the 90% interval of *Dangendorf et al.* (2024). This pre-1950 divergence follows from a design choice. We include only model representations that can be fit to observations and that describe the ongoing response to warming. Representing GMSL in earlier, colder periods would require more complete land-ice process models, which are outside our scope. We therefore treat the pre-1950 hindcast as qualitatively informative but do not draw quantitative conclusions from it. We restrict our validation claims to the 75-year, post-1950 period.

Figure 8b compares the rate of GMSL rise of our component model with observed rates as a function of GMST, with both rates estimated by the same local regression, a tricube kernel with a 30-yr half-width. The color of the model line shows its transient sensitivity along the 1957–2025 trajectory. The slope of the observed rate against GMST also agrees with the sensitivity of 3.4 mm yr^{-1} °C^{-1} found by *Rahmstorf* (2007) (Fig. 8b) nearly 20 years ago.

Together, the level, rate, and transient-sensitivity comparisons test our framework over a 75-year window (1950–2025) spanning about 1.0–1.1 °C of GMST change (11-yr means), without its having been calibrated on observations of GMSL itself. Over the satellite era its rates are 3.31 to 3.39 mm yr^{-1}, compared with 3.17 to 3.35 mm yr^{-1} in the four observational records over matched windows. Its 1950–2000 rate of 0.75 mm yr^{-1} is lower than those of the two tide-gauge reconstructions (1.01 and 1.02 mm yr^{-1}) and the 1.28 mm yr^{-1} of the IPCC AR6 compilation, by about the 0.44 mm yr^{-1} that the model-based terrestrial water storage subtracts over that period. Its transient sensitivity follows the observed increase in the rate of rise with warming.

This validation window is comparable in duration to the remaining forecast horizon, and in warming to the 2.7 °C scenario, which adds 1.4 °C between 2025 and 2100. The 1.8 °C scenario adds 0.4 °C, less than the warming across the validation window, although it takes GMST above any observed value, while the 3.6 °C scenario adds 2.6 °C, so its forecast extends furthest beyond what the hindcast tests. This validated envelope does not, however, extend to processes for which comparable historical constraints do not exist. WAIS, the largest source of forecast uncertainty, is therefore represented in this work by scenario-based storylines rather than a data-driven model (Section 2.2.2).

3.2 Component projections

We drive the calibrated model components with the IPCC AR6 GMST trajectories for the 1.8, 2.7, and 3.6 °C scenarios (Fig. 9a). Current policies imply a median warming of 2.7 °C by 2100, with emissions close to or slightly below those of the 2.7 °C scenario, although warming of 4 °C or more cannot be fully ruled out (*Hausfather*, 2025). The 1.8 and 3.6 °C scenarios bracket current policies. Projected levels are for 2100 relative to the study baseline year 2000, given as the posterior median with the 90% credible interval in brackets. We compare each component with the IPCC AR6 medium-confidence projections, which AR6 builds from processes that can be modeled with confidence. AR6 reports projections relative to the 1995–2014 mean (*Fox-Kemper et al.*, 2021). For these comparisons we subtract each ensemble member's own 1995–2014 mean, and we call the result the AR6 baseline.

The thermosteric contribution is 117 [93, 148], 163 [132, 202], and 209 [170, 257] mm at 1.8, 2.7, and 3.6 °C, respectively (Fig. 9b). On the AR6 baseline, our medians are 111, 157, and 202 mm, which are 0.78, 0.80, and 0.81 times the AR6 medians for thermal expansion of 142, 197, and 249 mm. On the same baseline, the upper layer contributes 51, 88, and 126 mm, the deep layer 48, 58, and 65 mm, and the ocean below 2000 m 11 mm in each scenario, so most of the difference between scenarios comes from the upper layer. The model has no explicit energy budget. Converted to ocean heat uptake with the AR6 expansion efficiency of 0.113 m YJ^{-1} (*Fox-Kemper et al.*, 2021), the projected rise between the 1995–2014 and 2081–2100 means is 0.79 to 0.82 times the heat uptake of the AR6 two-layer emulator in these scenarios (Section 1.8 of the Supplementary Material).

The Greenland Ice Sheet contributes 79 [41, 117], 102 [60, 147], and 129 [80, 180] mm at 1.8, 2.7, and 3.6 °C (Fig. 9c). On the AR6 baseline, our medians are 77, 100, and 126 mm, which are 1.39, 1.20, and 1.11 times the AR6 medians of 55, 83, and 114 mm. The calibrated ice-sheet model ensemble of *Aschwanden and Brinkerhoff* (2022) gives an independent comparison. Relative to 2008, when their simulations begin, it projects 80 [40,

140] mm under Representative Concentration Pathway (RCP) 2.6 and 110 [50, 180] mm under RCP4.5. Relative to the same year, we project 75 [37, 113] and 98 [56, 143] mm at 1.8 and 2.7 °C. Our Greenland discharge model is calibrated with the observed discharge record (*Mouginot et al.*, 2019), with two other mass-balance records withheld for validation (*Mankoff et al.*, 2021; *Otosaka et al.*, 2026). Our SMB emulator is trained on the MAR regional climate model alone. Among the regional climate models compared by *Glaude et al.* (2024), MAR is one of two that give about twice the SMB loss of RACMO under the same forcing.

We project glacier contributions of 119 [102, 141], 145 [121, 175], and 168 [138, 206] mm at 1.8, 2.7, and 3.6 °C (Fig. 9d). On the AR6 baseline, our medians are 116, 142, and 166 mm, which are 1.31, 1.16, and 1.05 times the AR6 medians of 89, 123, and 157 mm, so our glacier projection converges toward the AR6 median as warming increases. The GlacierMIP2 ensemble of 11 glacier models (*Marzeion et al.*, 2020) reports glacier mass loss over 2015–2100. Its medians and 90% intervals are 79 [23, 135] mm under RCP2.6, 119 [53, 185] mm under RCP4.5, and 159 [73, 245] mm under RCP8.5. Our glacier contribution in 2015 is 9.9 mm above the 2000 baseline, and over 2015–2100 we project medians of 109, 135, and 159 mm at 1.8, 2.7, and 3.6 °C. The first two lie within the GlacierMIP2 intervals for RCP2.6 and RCP4.5, respectively, and the third equals the RCP8.5 median. We model the rate of glacier mass loss as linear in GMST, following glacier model ensembles (*Marzeion et al.*, 2020; *Edwards et al.*, 2021), and calibrate it with the GlaMBIE record (*The GlaMBIE Team et al.*, 2025). The prior on the glacier sensitivity b_{gl} is weakly informative, and its posterior is set by the GlaMBIE record together with the pre-industrial balance constraint.

The combined Antarctic Peninsula and EAIS contribution stays near zero, with medians of −3.5 [−12.9, 4.5], −7.7 [−21.3, 4.0], and −11.4 [−29.0, 3.2] mm at 1.8, 2.7, and 3.6 °C (Fig. 9e). Negative values denote sea-level fall. All three 90% intervals span zero, so the sign of the net contribution is not resolved. The EAIS contribution becomes more negative across the three scenarios, from −12.7 to −23.6 mm, as warming increases snowfall. An Antarctic Peninsula contribution that rises from +9.1 to +11.9 mm partly offsets this. The EAIS projection adopts an SMB sensitivity derived from published accumulation scalings, $C_T = 60 \pm 20$ Gt yr^{-1} per °C of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013). The Peninsula sensitivity is calibrated with the IMBIE-3 Peninsula record (*Otosaka et al.*, 2026). It captures the combined response of SMB and ice discharge to warming, including glacier acceleration after ice-shelf collapse (*Scambos et al.*, 2004). West Antarctica is treated separately (Section 2.2.2).

3.3 Full model projections

We report the forecast first under the fast-WAIS storyline ($p(\mathrm{S1}) = 10\%$; Section 2.2.2). For the 1.8 °C scenario, this forecast shows a median GMSL rise of 0.59 [0.41, 1.58] m relative to 2000 (Fig. 10a). After referencing our projection to the AR6 baseline, this median is 32% above the IPCC AR6 medium-confidence median of 0.44 m. Under 2.7 °C warming, approximately the current trajectory (*Hausfather*, 2025), the median is 0.69 [0.49, 1.68] m, 21% above the AR6 median of 0.56 m. For the 3.6 °C scenario, the median reaches 0.78 [0.58, 1.77] m, 14% above the AR6 median of 0.68 m. For reference, the 4.4 °C scenario yields a median of 0.87 [0.63, 1.86] m.

Relative to pre-industrial GMSL (adding ∼0.15 m, the 0.14 m of observed rise from 1900 to 2000 (*Frederikse et al.*, 2020; *Dangendorf et al.*, 2024) plus ∼0.01 m from the 1850–1900 mean to 1900 (*Kopp et al.*, 2016; *Fox-Kemper et al.*, 2021)), these forecasts correspond to medians of 0.74–0.93 m of GMSL rise across the 1.8–3.6 °C policy-relevant scenarios. The probability of exceeding 1 m relative to pre-industrial by 2100 is 22% under 1.8 °C of warming, 28% under 2.7 °C, and 38% under 3.6 °C. The probability of exceeding 2 m relative to pre-industrial by 2100 ranges from 3 to 4% across the 1.8–3.6 °C scenarios and is lowest at 1.8 °C. Under the slow-WAIS storyline, the probability of exceeding 1 m relative to pre-industrial by 2100 falls to 2.5% at 1.8 °C, 3.1% at 2.7 °C, and 4.7% at 3.6 °C, and the probability of exceeding 2 m stays below 1% across the 1.8–3.6 °C scenarios (Fig. 10b). For comparison, our slow-WAIS forecast for 2.7 °C of warming is 87 mm below the extrapolation of the observed GMSL trend, well within the 90% interval of the extrapolation, 0.40 to 0.85 m. The two are expected to be close if warming continues along its current trend and WAIS, which contributed 9% of observed GMSL rise from 1993 to 2023, continues to lose mass at its current rate.

Fig. 11a shows the mean 21st-century contribution of each component. Under the fast-WAIS storyline, WAIS governs the upper tail of the forecast distribution in every warming scenario (Fig. 10).

The distribution of the uncertainties shows a clear separation between the predictable components and those that are currently unpredictable (Fig. 11b). Under the fast-WAIS storyline, WAIS accounts for more than 90% of the variance of 21st-century sea-level rise, and for 82 to 87% of the summed widths of the component 90% credible intervals. The thermosteric, glacier, and Greenland contributions, whose sensitivities to GMST are calibrated or, for Greenland surface mass balance, emulated, contribute 0.7 to 2% to the total variance and 13 to 18% to the summed interval widths, with their share growing as warming increases. As a result, the 90% uncertainty range of 1.20 m for 2.7 °C is 6.2 times the spread across warming scenario medians (0.19 m between 1.8 and 3.6 °C). This picture shifts under the slow-WAIS storyline. WAIS still accounts for 90 to 96% of the variance, but the three calibrated components set 38 to 48% of the summed interval widths, up from 13 to 18% (right column of Fig. 11b). Because the WAIS forecast does not depend on GMST before 2100, sea-level risk separates into two dimensions, one reducible by emissions cuts and one set by WAIS, each addressed by a distinct class of societal response.

4 Discussion

4.1 Data-driven forecasts as part of a hierarchy

The SPECTRUM framework occupies an intermediate level of a hierarchy of sea-level models ordered by complexity, so its results can be compared with both simpler and more complex models. The simplest models extrapolate the observed trends. Extending our quadratic fit to the satellite altimetry record of GMSL to 2100, following *Nerem et al.* (2022) and *Hamlington et al.* (2024), who stop at 2050, gives 0.63 [0.40, 0.85] m of rise relative to 2000. At the local scale, trends and accelerations at 222 tide gauges over 1970–2023, extrapolated to 2050, are consistent with the AR6 medium-confidence projections within the likely range

at 96% of locations (*Wang et al.*, 2025). Next in complexity are the semi-empirical models, which relate the rate of global mean sea-level rise to GMST. Applying the semi-empirical relation of *Rahmstorf* (2007) to the 2.7 °C scenario gives 0.73 [0.67, 0.82] m relative to 2000, where the interval reflects only the spread of warming paths. Our slow-WAIS forecast (p(S1) = 90%) gives 0.54 [0.42, 0.73] m at 2.7 °C (Fig. 10a). Its median is 87 mm less than the extrapolation of GMSL and 0.19 m less than the semi-empirical model. The agreement to within 0.19 m between the slow-WAIS forecast at 2.7 °C, which approximates the current warming trend, and the simpler approaches is a consistency check on the forecast.

At the complex end of the hierarchy are the process-model assessments of IPCC AR6 (*Fox-Kemper et al.*, 2021), which report two sets of projections. The medium-confidence set is built from processes that can be modeled with confidence, while the low-confidence set admits ice-sheet processes such as marine ice cliff instability along with expert judgment. In SPECTRUM, p(S1) moves the forecast from the medium-confidence set toward the low-confidence set. To compare with AR6, we shift our forecast to the AR6 1995–2014 baseline, which lowers each value by 13 mm. On that baseline, our slow-WAIS forecast gives 0.44 [0.33, 0.62] m at 1.8 °C, 0.53 [0.41, 0.72] m at 2.7 °C, and 0.62 [0.49, 0.83] m at 3.6 °C. The AR6 medium-confidence medians are 0.44, 0.56, and 0.68 m. Our slow-WAIS median equals the AR6 median at 1.8 °C and is 5 and 8% lower at 2.7 and 3.6 °C, whereas the fast-WAIS medians are 14 to 32% higher. Our slow-WAIS 95th percentiles are less than the AR6 medium-confidence 95th percentiles at all three warming levels (0.62, 0.72, and 0.83 m, compared with 0.78, 0.95, and 1.11 m). At all three warming levels, the difference between our fast-WAIS forecast and the AR6 medium-confidence projections is a difference in the weight given to rapid WAIS mass loss. The thermosteric, glacier, and Greenland contributions together differ from their AR6 medians by +18, −4, and −26 mm at 1.8, 2.7, and 3.6 °C, compared with differences of 141, 116, and 94 mm in the total. Our fast-WAIS forecast is greater than the AR6 low-confidence projections at 1.8 and 2.7 °C and less at 4.4 °C, the three scenarios for which AR6 provides them. At 2.7 °C, our median of 0.67 m and 95th percentile of 1.67 m are greater than the low-confidence values of 0.57 and 1.18 m. At 4.4 °C, our median of 0.86 m and 95th percentile of 1.85 m are less than the low-confidence values of 0.88 and 2.28 m.

4.2 Completeness of the uncertainty budget

Our intervals include most sources of uncertainty, but not alternative model forms for the calibrated components. *Aschwanden et al.* (2021) set aside the emissions scenario and separate the remaining uncertainty in an ice-sheet forecast into four parts. These are the model structure, the initial state, the model parameters, and random variability in the forcing. They argue that the ISMIP6 ensemble sampled model structure but not parameters, and that for Greenland the parametric spread could be the larger of the two. SPECTRUM samples the parameters of every calibrated component. WAIS, the one component whose uncertainty is dominated by model structure, is represented instead by a mixture of scenarios. Each warming scenario allows a range of GMST paths, and each ensemble member follows one path shared by all components. Errors in the initial state and year-to-year variability are absorbed into the calibrated error models rather than represented separately. The intervals do not include structural uncertainty in the calibrated components, that is, alternative functional

forms for each component. This omission matters most under the slow-WAIS storyline, in which the thermosteric, glacier, and Greenland contributions account for 38 to 48% of the summed widths of the 90% credible intervals of the four largest components. Under the fast-WAIS storyline it matters little, because WAIS accounts for more than 90% of the variance. We tested one alternative functional form directly. Our glacier model assumes that glaciers never shrink toward a size in balance with the warmer climate, which is the limit of an infinitely long response time (Eq. 17). Of course, glaciers do shrink, and their loss rate declines as they do, over decades to centuries (*Roe et al.*, 2021). A model that includes this adjustment fits the glacier record as well as ours when the response time is a century or more. Averaged over response times of 20 to 200 years, it gives about 30 mm less glacier sea-level rise from 2000 to 2100 at each warming level, and 11 to 13 mm less if response times up to millennia are equally plausible. Our glacier projections lie within the range of the process model of *Rounce et al.* (2023), so the bias from neglecting this adjustment is likely toward the smaller value. Three further tests perturb inputs to the Greenland discharge model. Moving the center of the sampled Arctic amplification factor from 2.5 to either end of the observed Greenland-to-global warming ratio (2.0 and 2.7) moves the 2100 median by 5.1 mm or less. Shifting the 200–500 m ocean-forcing layer 100 m shallower or deeper moves it by 2.5 mm or less, and extending the range of ocean response times τ_{go} from 1–20 yr to 1–40 yr moves it by 2.6 mm or less. No test changes the width of the 90% credible interval by more than 5.9 mm.

We do not calculate the integral over model structure itself, in which a projection is averaged over a distribution across model structures (*Aschwanden et al.*, 2021). Assigning weights that sum to one to the models that have been built would treat each model as an independent sample of the ways the system might be represented. In fact those models describe the same system, rely on many of the same assumptions, and are tested against many of the same observations. The ISMIP6 emulator of *Edwards et al.* (2021), which underlies the AR6 medium-confidence Antarctic projections, is an example. It treats each multi-model ensemble as the output of a single stochastic simulator and uses all models, because observations cannot yet identify which responds most realistically. A random-effects test by its authors found that dependence among the models has little effect on the projections at 2100. Our two WAIS storylines bracket the outcome rather than estimate that distribution. SPECTRUM does implement the conditioning step of *Aschwanden et al.* (2021), in which a prior over projections is filtered by observations, for every component except WAIS. Each component is calibrated with its own observational record, and observed GMSL is withheld from all of them. The hindcast is therefore a closure test, which asks whether the separately fitted components add up to the observed total. Over 1950–2000 the summed components rise at 0.74 mm yr^{-1}, compared with 1.02 and 1.01 mm yr^{-1} in the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions and 1.28 mm yr^{-1} in the IPCC AR6 compilation. *Lin et al.* (2026) take the alternative approach for Antarctica. They emulate the ISMIP6-2300 ensemble, whose ice-model choices were documented but not sampled systematically, and place a uniform prior over 18 ice-dynamical features and the choice of climate model. In the raw ISMIP6 simulations emulated by *Lin et al.* (2026), choices made within the ice-sheet models account for 64% of the 2015–2100 variance, about four times the share of the forcing. In their drift-corrected simulations the ranking reverses, with 23% from the model choices compared with 48% from the forcing. Lin et al. then update the probabilities

with the observed mass-loss rate. The observations constrain the sliding law, but for most other choices the result remains strongly dependent on the prior. A two-decade record says little about processes that barely affect the recent trend but strongly affect the century-scale response. *Lin et al.* (2026) describe this residual as strong ambiguity and bound it with probability boxes, which give the range of probabilities across a set of plausible priors. The same difficulty leads us to represent WAIS with a scenario mixture rather than a single distribution.

4.3 Two dimensions of sea-level risk

Sea-level risk in 2100 has two dimensions. The first comes from the thermosteric, glacier, and Greenland contributions, which respond to warming and can be reduced by cutting emissions. The second comes from WAIS, which in our forecast does not respond to emissions this century. We calibrate the warming sensitivities of the thermosteric, glacier, and Greenland components with decades of observations and validate the fits with decades of observations withheld from the calibration. Their collective contribution shifts systematically across warming trajectories and is reducible by emissions cuts (Fig. 11a). Under the slow-WAIS storyline, the 90% credible interval of the forecast spans 0.30 m at 1.8 °C, 0.31 m at 2.7 °C, and 0.33 m at 3.6 °C. The fast-WAIS intervals, which admit rapid, feedback-driven WAIS mass loss, are 3.6 to 4.0 times wider (Fig. 10). If WAIS loses mass slowly, emissions policy is the main lever for managing sea-level risk. Under the slow-WAIS storyline, warming of 3.6 °C rather than 1.8 °C adds 0.19 m to the median rise by 2100. That added rise adds US$25 billion per year in coastal adaptation costs.

The second dimension is the width of the upper tail, which changes little with emissions over the 21st century. Under the fast-WAIS storyline, WAIS accounts for more than 90% of the forecast variance, and in our forecast it does not depend on the emissions pathway before 2100. The tail of the forecast distribution holds the outcomes with the largest human and economic consequences. It is controlled almost entirely by whether WAIS follows modern loss trends or transitions to more rapid retreat (*Turney et al.*, 2020; *Dutton et al.*, 2015; *Graham et al.*, 2022). Under the fast-WAIS storyline, emissions reductions shift the whole distribution downward, lowering the median and the 95th percentile by similar amounts, but do not narrow the upper tail, whose spread above the median (0.99 and 0.98 m at 1.8 and 3.6 °C) is set by WAIS. The probability of exceeding 2 m relative to pre-industrial is 3% at 1.8 °C and 4% at 3.6 °C. Even at 1.8 °C, emissions reductions alone therefore cannot remove the high-risk outcomes. This 3% rests on $p(\mathrm{S1}) = 10\%$ and on an S2 95th percentile set near the AR6 low-confidence Antarctic value, which combines structured expert judgment with simulations that include marine ice cliff instability. Emerging research on processes absent from current ice-sheet models points toward mechanisms that could raise those values (Section 2.2.2).

The costs to coastal communities show the gap between the fast-WAIS and slow-WAIS storylines (Figs. 10 and 11a). Risk-averse planning for long-lived infrastructure needs information on the upper tail of the distribution (*van de Wal et al.*, 2022); here we use the 95th percentile as one such design threshold (Fig. 10b). Under the current-policy warming trajectory (2.7 °C), this threshold is 0.95 m higher by 2100 under the fast-WAIS storyline than under the slow-WAIS storyline (1.68 m compared with 0.73 m). This difference trans-

lates into US$128 billion per year in coastal adaptation costs and US$12.6 trillion per year in coastal flood damages without further adaptation. For intuition, the fast-WAIS 95th percentile implies more than seven times the sea-level rise of the past 125 years within the next 75 years, and the slow-WAIS 95th percentile about three times.

Scientific research is necessary for providing actionable information for coastal adaptation. Further research may add to the evidence for rapid WAIS mass loss rather than build confidence in slow loss, because key processes remain missing from or crudely represented in models (*Fricker et al.*, 2025). Under the fast-WAIS storyline at 2.7 °C, there is a 5% probability of less than 0.49 m and a 5% probability of more than 1.68 m of GMSL rise by 2100. While the range of outcomes remains this wide, decisions on coastal planning, adaptation, and managed retreat that affect millions of lives and cost hundreds of billions of dollars per year will be made under large uncertainty. The extent of these uncertainties in terms of lives, property, and ecosystems raises the question of whether action could make WAIS more stable, and how much harm to people and the environment that could prevent. Research directions to address this question have been proposed (*MacAyeal et al.*, 2024), and initial models of glacier interventions aimed at stabilizing WAIS show promise (*Meyer et al.*, 2026; *Miller et al.*, 2026). In the context of this study, stabilizing WAIS means increasing $p(\mathrm{S1})$ and reducing the 95th percentile of S2, the scenario that admits rapid WAIS mass loss. As with any nascent research, there are many unresolved questions about the feasibility, unintended consequences, and governance of these interventions (*Siegert et al.*, 2025). There are risks in intervening to slow the loss of WAIS. The logistical challenges will also be large, but these risks and challenges should be weighed against the risks and challenges of adapting to sea-level rise. There is no response to sea-level rise that comes with zero risk and zero cost.

5 Conclusions

Sea-level rise this century has two parts that call for different responses. The first is the combined contribution of thermal expansion, glaciers, and the GrIS. Its sensitivity to warming is calibrated with observations or emulated from a regional climate model, and emissions cuts reduce it. The second is WAIS. Observations cannot yet tell how its mass loss will accelerate, and it dominates the uncertainty in the forecast.

SPECTRUM represents each contributor with a simple physical model, including a new ocean-forced model of Greenland outlet-glacier discharge, and represents WAIS with two scenarios, one of which extends the observed trend. With observed GMSL withheld from every calibration, the summed components account for 89–104% of the satellite-altimetry rate over three periods since 1993. Their 1950–2000 rate is 0.3–0.5 $\mathrm{mm\,yr^{-1}}$ lower than observed, by about the contribution of the model-based terrestrial water storage. The 1950–2025 period spans about 1.0–1.1 °C of warming, comparable to the 1.4 °C that the 2.7 °C (SSP2-4.5) scenario adds after 2025 and less than half of the 2.6 °C that the 3.6 °C (SSP3-7.0) scenario adds. Under the slow-WAIS storyline ($p(\mathrm{S1}) = 90\%$), median GMSL rise over 2000–2100 is 0.45, 0.54, and 0.64 m at 1.8, 2.7, and 3.6 °C, with a 90% credible interval of 0.42 to 0.73 m at 2.7 °C. The median at 2.7 °C lies within 0.19 m of both the extrapolated satellite-era trend and the semi-empirical model, and on the AR6 baseline the slow-WAIS medians are equal to or up to 8% below the AR6 medium-confidence medians. Under the

fast-WAIS storyline (p(S1) = 10%), the median at 2.7 °C is 0.69 m, with a 90% credible interval of 0.49 to 1.68 m. Even with slow loss assigned 90% probability, WAIS accounts for 90 to 96% of the forecast variance at 2100, mostly through the possibility of rapid loss. Under the slow-WAIS storyline, the thermosteric, glacier, and Greenland contributions set 38 to 48% of the summed widths of the component 90% credible intervals. Their intervals do not include alternative model forms.

Emissions cuts lower the median rise. Under the slow-WAIS storyline, holding warming to 1.8 °C rather than 3.6 °C lowers the median by 0.19 m. We assume that WAIS mass loss before 2100 is largely committed and independent of warming (*Naughten et al.*, 2023; *Coulon et al.*, 2024; *Rosier et al.*, 2025). Under the fast-WAIS storyline, the same cut therefore leaves the spread above the median near 1 m, and the probability of exceeding 2 m relative to pre-industrial by 2100 falls from 4 to 3%. These tail values rest on p(S1) and on the 95th percentile adopted for the fast scenario. Beyond 2100, emissions matter for WAIS as well, because 21st-century warming largely sets the long-term Antarctic commitment (*Lowry et al.*, 2021; *Seroussi et al.*, 2024). Better observations and models could tell planners which outcome they face, although they may add to the evidence for rapid loss rather than rule it out (*Fricker et al.*, 2025). Whether interventions could slow WAIS mass loss, and so act on the tail, is an open question. Emissions cuts shift the distribution of sea-level rise, and the rate of WAIS mass loss sets its upper tail.

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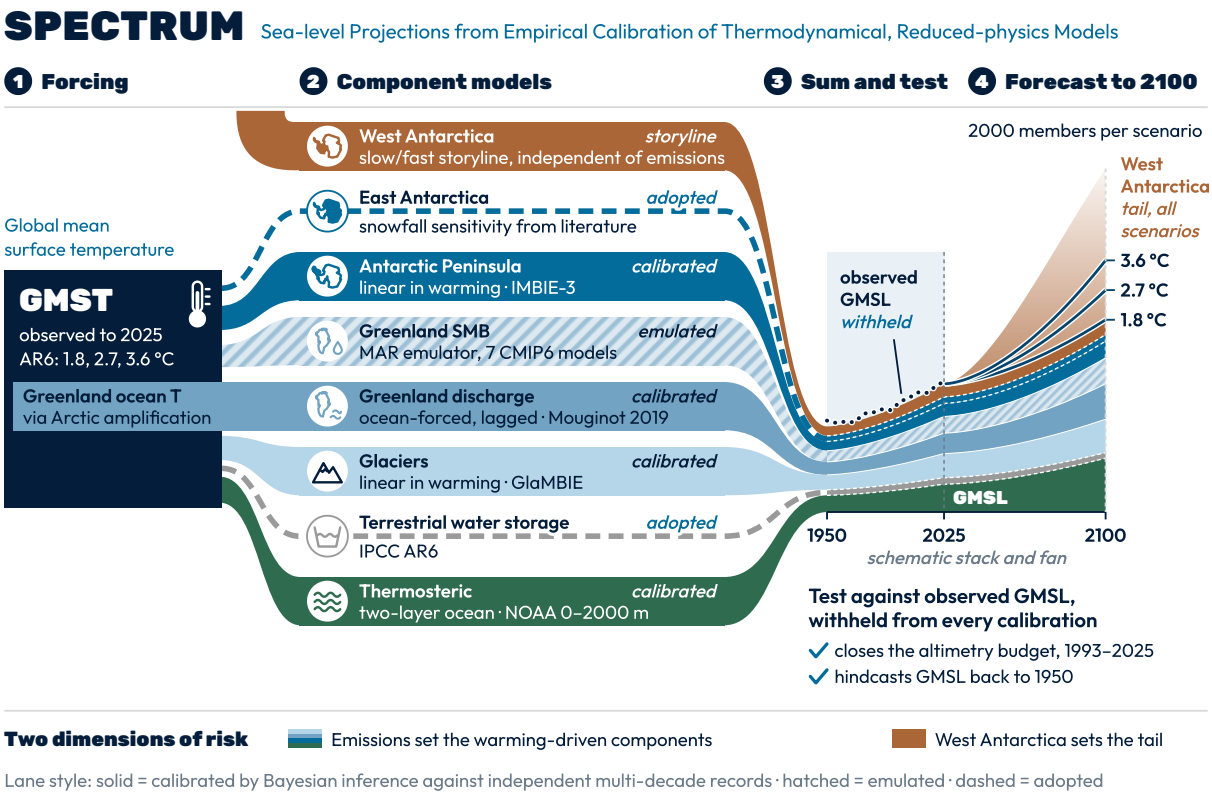


Figure 1: **The SPECTRUM framework.** (1) Global mean surface temperature (GMST), observed to 2025 and following the AR6 1.8, 2.7, and 3.6 °C paths, forces each component. Greenland discharge is forced by ocean temperature around Greenland, derived from GMST through Arctic amplification. (2) Each contributor has its own model. Line style marks how the model is constrained: solid lanes are calibrated by Bayesian inference against independent multi-decade records, the hatched lane is emulated, and dashed lanes are adopted from published sensitivities and assessments. West Antarctica (brown) follows a slow/fast storyline that is independent of emissions this century. (3) The components are summed and tested against observed GMSL, which is withheld from every calibration. (4) The forecast to 2100 uses 2000 members under each warming level. Emissions set the warming-driven components, and West Antarctica sets the tail in all scenarios. The stacked areas and fan are schematic.

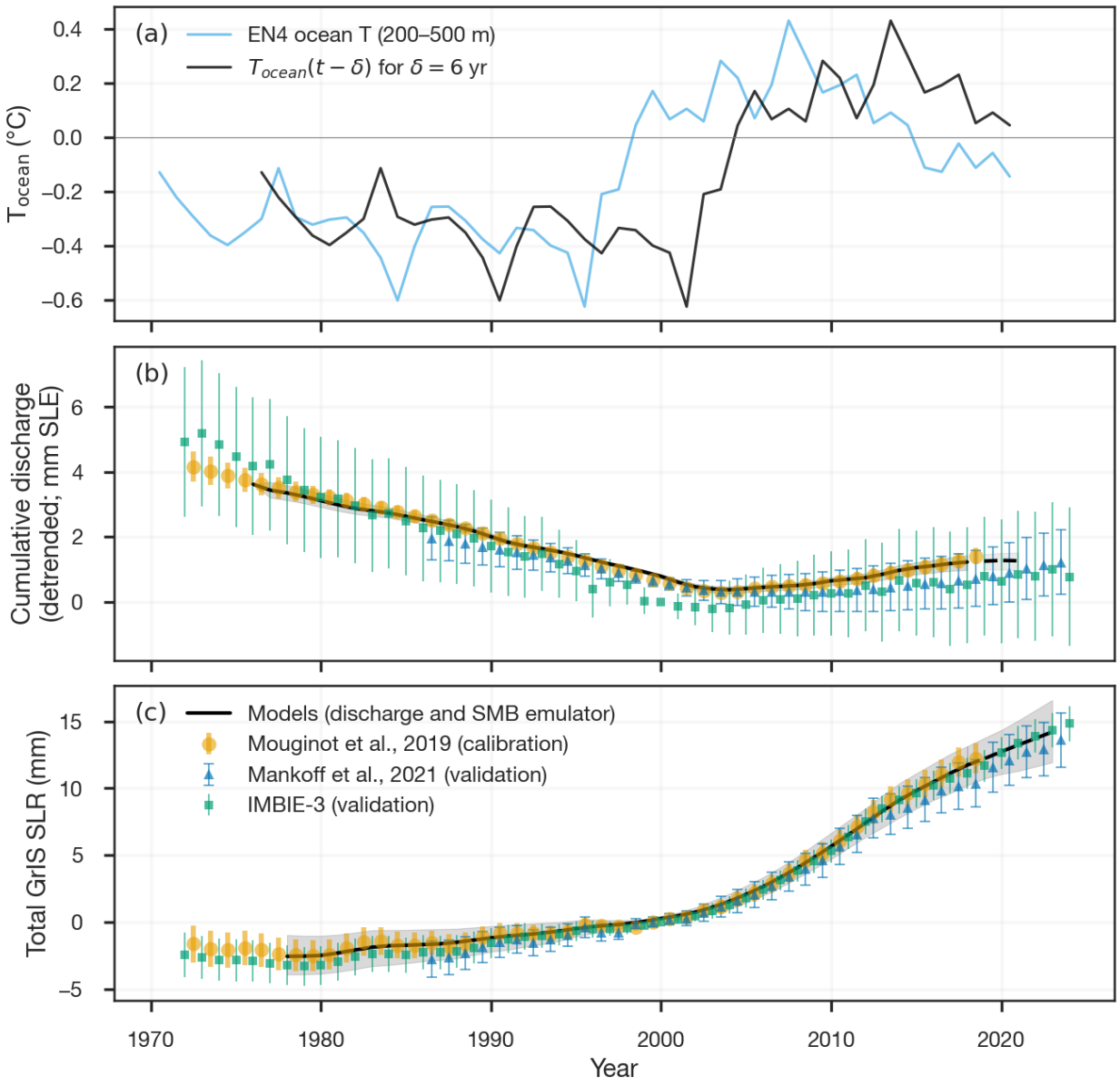


Figure 2: **Calibration and validation of the Greenland discharge model. (a)** Subsurface ocean temperature anomaly relative to 1995–2005 (from EN4, averaged over depths of 200–500 m and over 58–80°N, 75–5°W, with annual resolution; blue) and the same signal delayed by $\delta = 6$ yr, $T_{\mathrm{ocean}}(t-\delta)$ (black). **(b)** Detrended cumulative Greenland ice discharge, showing the model fit (black) with its 90% credible interval, compared with the calibration data (*Mouginot et al.*, 2019) (orange) and the time series of *Mankoff et al.* (2021) (blue) and IMBIE-3 (*Otosaka et al.*, 2026) (green) used for validation. Each record is detrended by the background rate r_0 from a fit of the same model to that record. **(c)** Total Greenland contribution to GMSL, the sum of the modeled SMB and discharge contributions (black, with 90% credible interval). SMB comes from a statistical emulator of the MAR regional climate model. Dashed lines continue the models past the end of the *Mouginot et al.* (2019) record. Error bars on the data are 90% intervals.

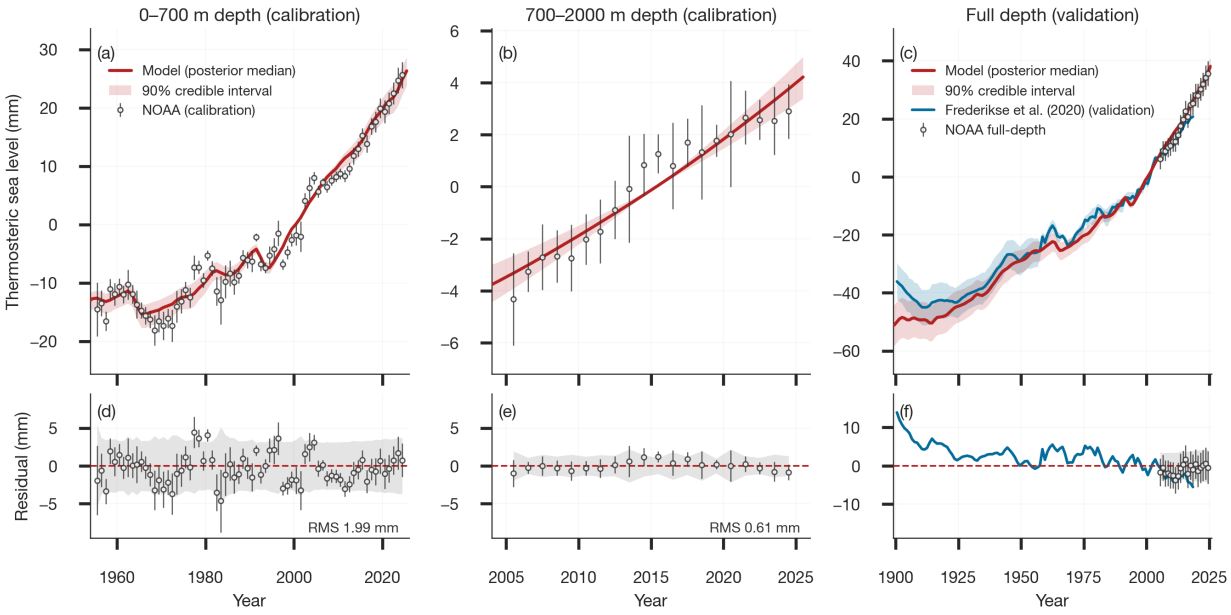


Figure 3: **Two-layer thermosteric model fit and validation.** (a) Posterior median of the upper layer model H_{u} (Eq. 8) and 90% credible interval fitted to observations at 0–700 m depth (circles with error bars; from NOAA (*Levitus et al.*, 2012)). (b) Same as in (a) but for the 700–2000 m data, the difference between the NOAA 0–2000 m and 0–700 m records, and the deep layer $b_{\mathrm{d}}S_{\mathrm{d}}$, both relative to their 2005–2024 means. (c) Validation of the full-depth model H_{full} (Eq. 12) against the results from *Frederikse et al.* (2020) (blue). Also shown for reference is a full-depth estimate built from the NOAA 0–700 m and 0–2000 m records and the constant rate of expansion below 2000 m (circles). Error bars on the observations are 90% intervals of their formal and structural errors, and in (c) and (f) they also include $\sigma_{\mathrm{extra,u}}$, $\sigma_{\mathrm{extra,d}}$, and the uncertainty of the rate below 2000 m. (d–f) Residuals corresponding to (a)–(c), with the gray band showing the 90% interval of the total point error, which includes $\sigma_{\mathrm{extra,u}}$ in (d) and $\sigma_{\mathrm{extra,d}}$ in (e), and the RMS residual reported in each calibration panel.

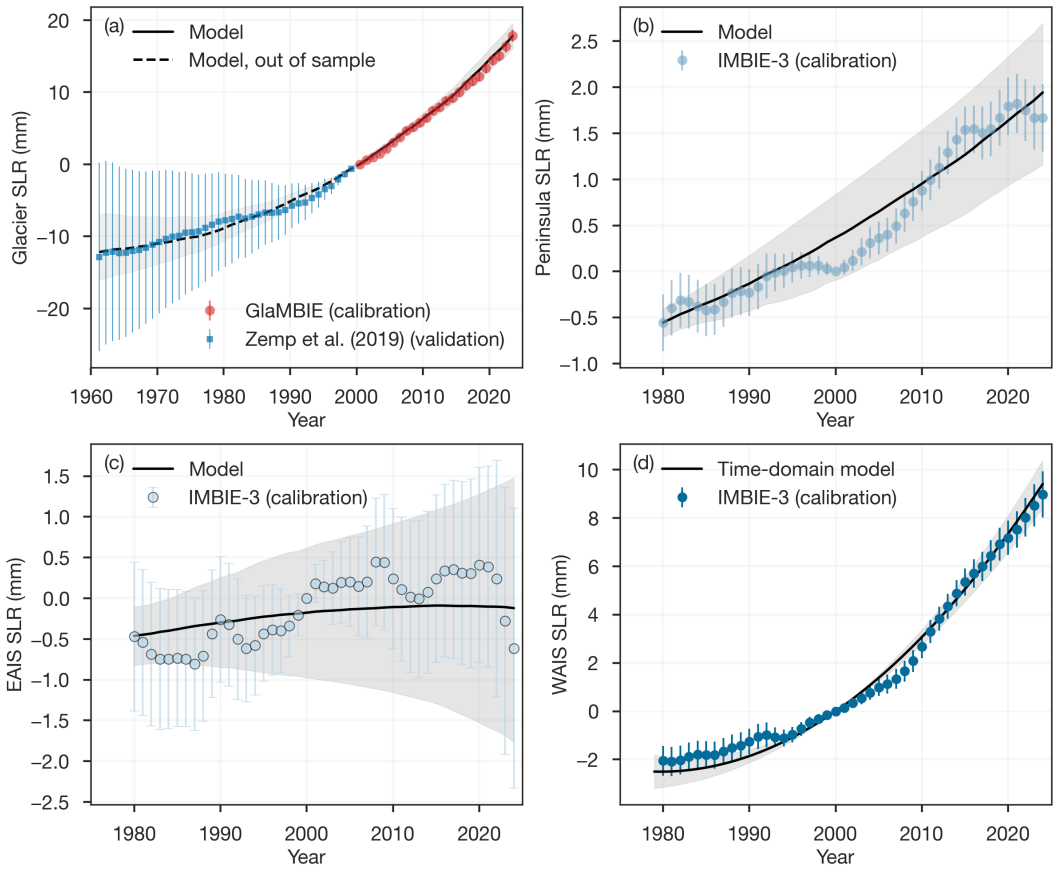


Figure 4: **Per-component calibration fits. (a)** Glaciers, **(b)** Antarctic Peninsula, and **(c)** EAIS observed cumulative contributions to GMSL (circles with 90% error bars) are shown with the respective posterior median models (black lines) and their 90% credible intervals (gray shading). **(d)** The WAIS cumulative contribution is shown with a quadratic fit in time rather than in temperature (black line) and its 90% credible interval (gray shading). In (a) the glacier model is also run back to 1961 (dashed line and lighter shading, outside the calibration period) and compared with the validation record of *Zemp et al.* (2019) (squares), whose cumulative values are matched to the model at the end of 1999. Data are from *The GlaMBIE Team et al.* (2025) (calibration) and *Zemp et al.* (2019) (validation) for glaciers and *Otosaka et al.* (2026) for the ice sheets.

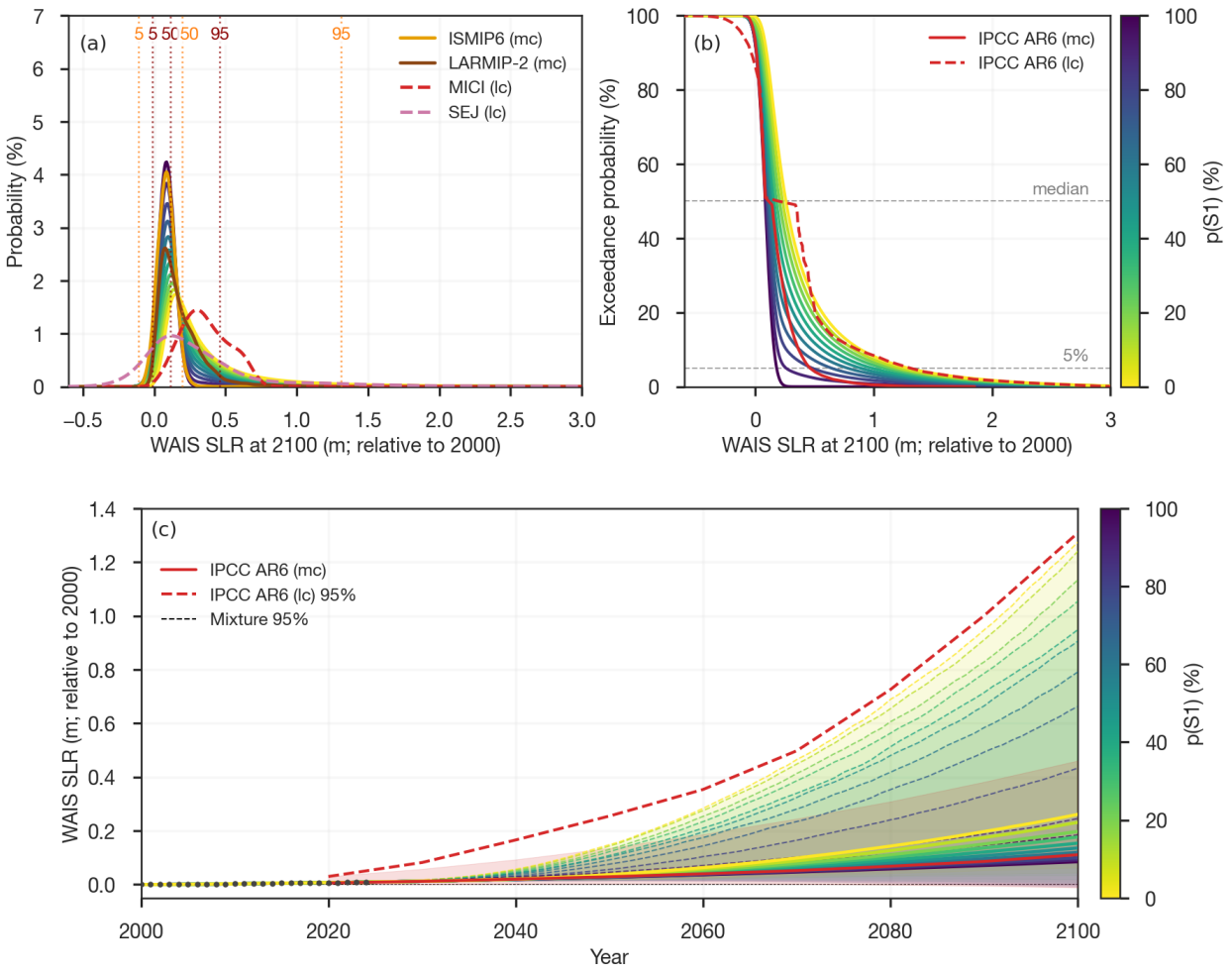


Figure 5: **WAIS scenario mixture model. (a)** Probability density of the WAIS contribution to GMSL at 2100 (relative to 2000) for an 11-member sweep of the mixture weight p(S1) from 100% (S1, continued observed trend) to 0% (S2, accelerated loss), colored by weight (colorbar). Also shown are the IPCC AR6 model ensembles for the whole Antarctic Ice Sheet that underlie its medium-confidence projections at 2.7 °C (mc; ISMIP6 emulator, *Edwards et al.*, 2021; LARMIP-2 ensemble, *Levermann et al.*, 2020) and its low-confidence projections at 4.4 °C (lc; marine ice cliff instability, MICI, *DeConto et al.*, 2021; structured expert judgment, SEJ, *Bamber et al.*, 2019). Dotted vertical lines mark the 5th, 50th, and 95th percentiles of the AR6 medium-confidence (red) and low-confidence (orange) projections. **(b)** Corresponding exceedance probability curves for the same p(S1) sweep, with the IPCC AR6 medium-confidence 2.7 °C (red solid line) and low-confidence 4.4 °C (red dashed line) exceedance curves for the Antarctic Ice Sheet overlaid for reference. **(c)** Time series for the p(S1) sweep and the IPCC AR6 medium-confidence 2.7 °C projection (red solid line with 5–95% range shaded) and low-confidence 4.4 °C 95th percentile (red dashed line). The thin dashed lines show the 95th percentile for the correspondingly colored p(S1), and the circles show the IMBIE-3 record with 90% intervals. The AR6 curves in all panels are on the AR6 1995–2014 baseline, which differs from the 2000 baseline of our curves by a few millimeters, and the LARMIP-2 ensemble is its total of ice dynamics and surface mass balance.

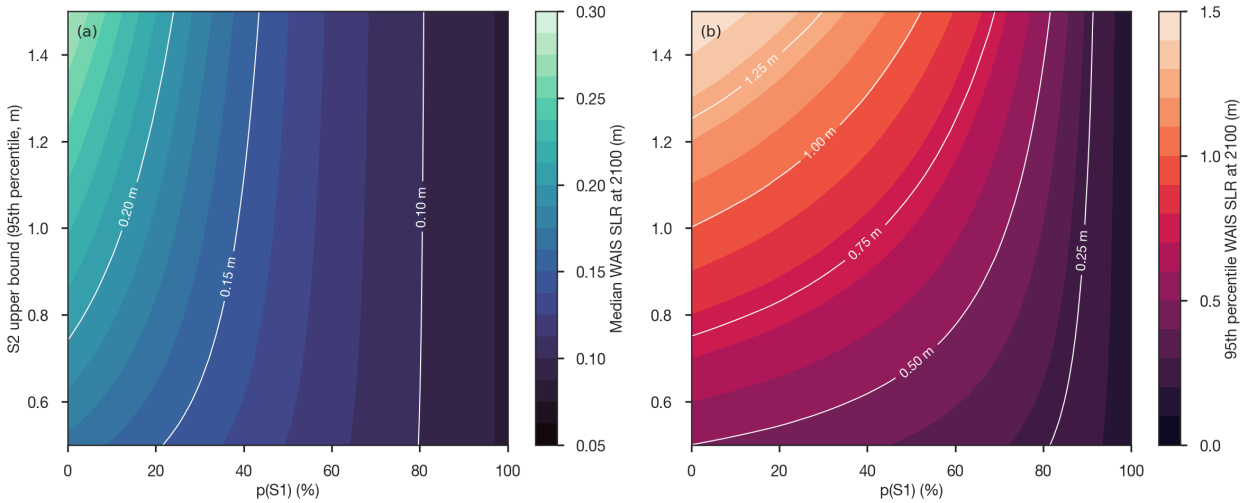


Figure 6: **WAIS contribution to GMSL at 2100 as a function of the two mixture parameters. (a)** Median and **(b)** 95th percentile of the S1/S2 mixture (relative to 2000) across the S1 probability, $p(\text{S1})$, and the S2 95th percentile at 2100.

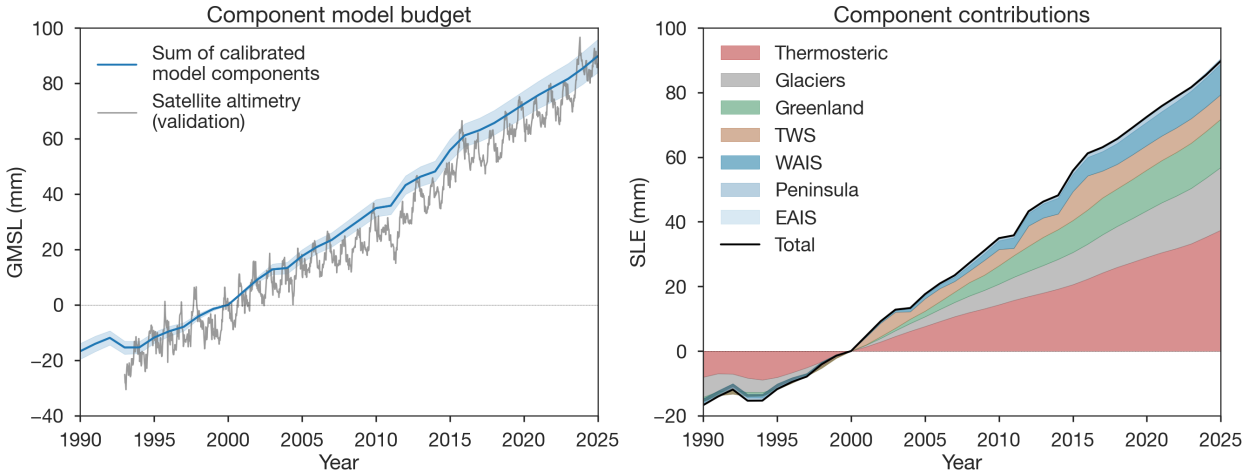


Figure 7: **Budget closure (validation): sum of model components versus observed GMSL. (a)** Sum of components (blue, with shading for its 90% credible interval) and NASA satellite altimetry (gray) (*Beckley et al.*, 2017; *Hamlington et al.*, 2020). **(b)** Stacked per-component contributions and their sum in black. Terrestrial water storage follows *Frederikse et al.* (2020) and is held at its 2018 value thereafter.

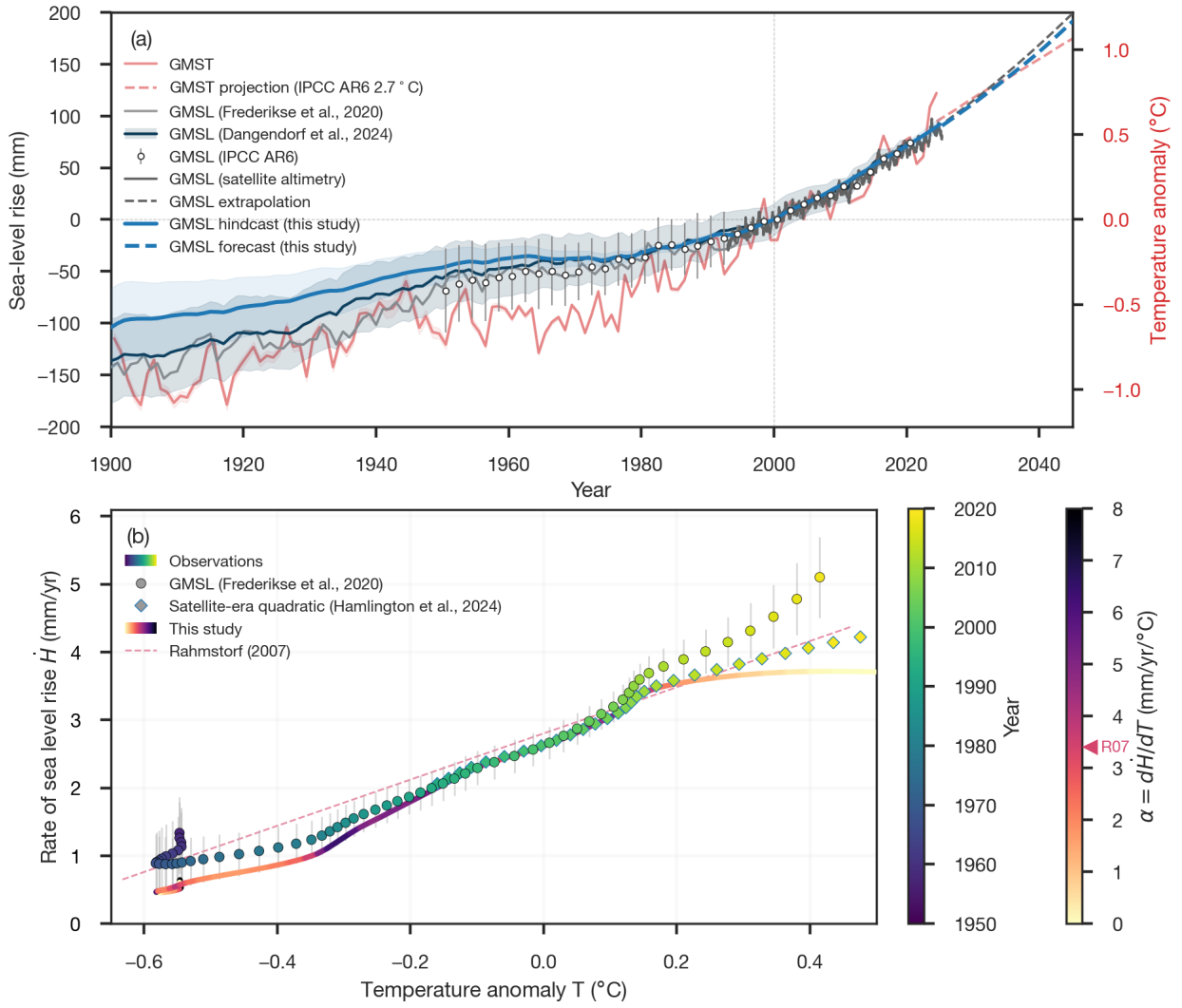


Figure 8: **Global mean sea-level rise and global mean surface temperature. (a)** GMSL reconstructions from *Frederikse et al.* (2020) (light gray) and *Dangendorf et al.* (2024) (navy, with 90% uncertainty range shaded), the IPCC AR6 observational compilation (*Fox-Kemper et al.*, 2021) (white circles with 5–95% range bars), and NASA satellite altimetry (*Beckley et al.*, 2017) (mid gray, with the satellite-era quadratic of *Hamlington et al.* (2024) extrapolated as a dashed line) show sea level rising with warming. The hindcast of this study is shown in solid blue with shading for its 90% credible interval, and the median forecast for the 2.7 °C scenario in dashed blue. Before 2000 the hindcast includes the prescribed contribution of uncharted glaciers (*Parkes and Marzeion*, 2018). GMST (red) (*Rohde and Hausfather*, 2020), relative to its 1995–2005 mean and continued by the AR6 median path of the 2.7 °C scenario (dashed red), is referenced to the right-hand axis. **(b)** Rate of GMSL rise ($\dot{H} = dH/dt$) versus GMST anomaly, T. Observed values are shown in circles (*Frederikse et al.*, 2020), colored by year, with 90% error bars that account for the correlated errors of a cumulative record, and diamonds show the rate of the satellite-era quadratic of *Hamlington et al.* (2024). Both the observed and the model rates include terrestrial water storage. The component model of this study, the median of the summed component ensemble without blending, is a line colored by instantaneous transient sensitivity $\alpha = d\dot{H}/dT$, with rates from the same local regression as the observations, a tricube kernel with a 30-yr half-width.

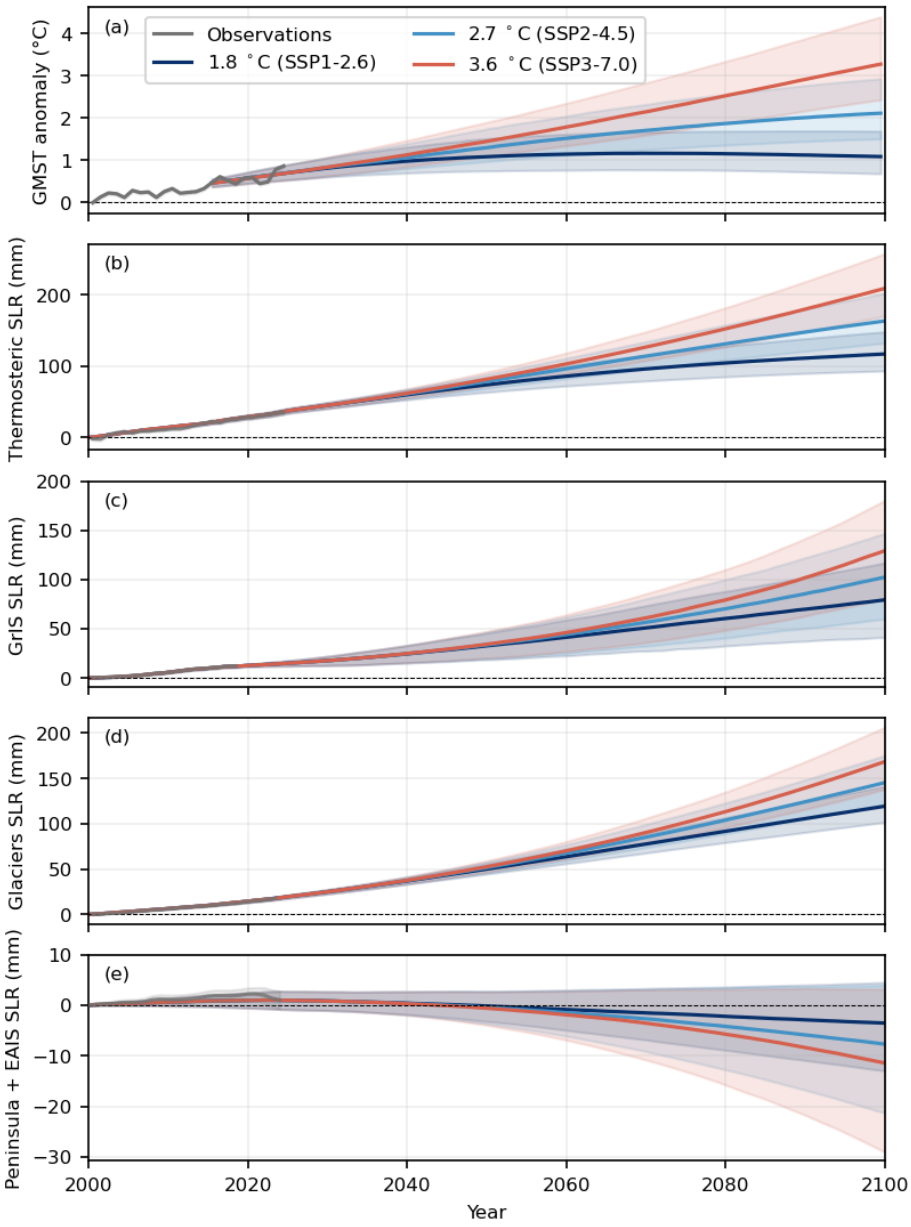


Figure 9: **Component-level forecasts under three warming scenarios.** (a) Observed global mean surface temperature (GMST) anomaly (gray) relative to 2000, with the IPCC AR6 GMST trajectories for the 1.8, 2.7, and 3.6 °C scenarios and their 5–95% ranges (shaded). The curves show warming relative to 2000. The legend labels each scenario by its AR6 central-estimate warming over 2081–2100 relative to the pre-industrial (1850–1900) mean. (b) Thermosteric sea-level contribution relative to 2000, with observations (gray) and the forecast median with its 90% credible interval from the full-depth two-layer model driven by the corresponding GMST trajectory. (c) Greenland Ice Sheet, (d) glaciers, and (e) the sum of the Antarctic Peninsula and East Antarctic Ice Sheet (EAIS) contributions, each shown as in (b).

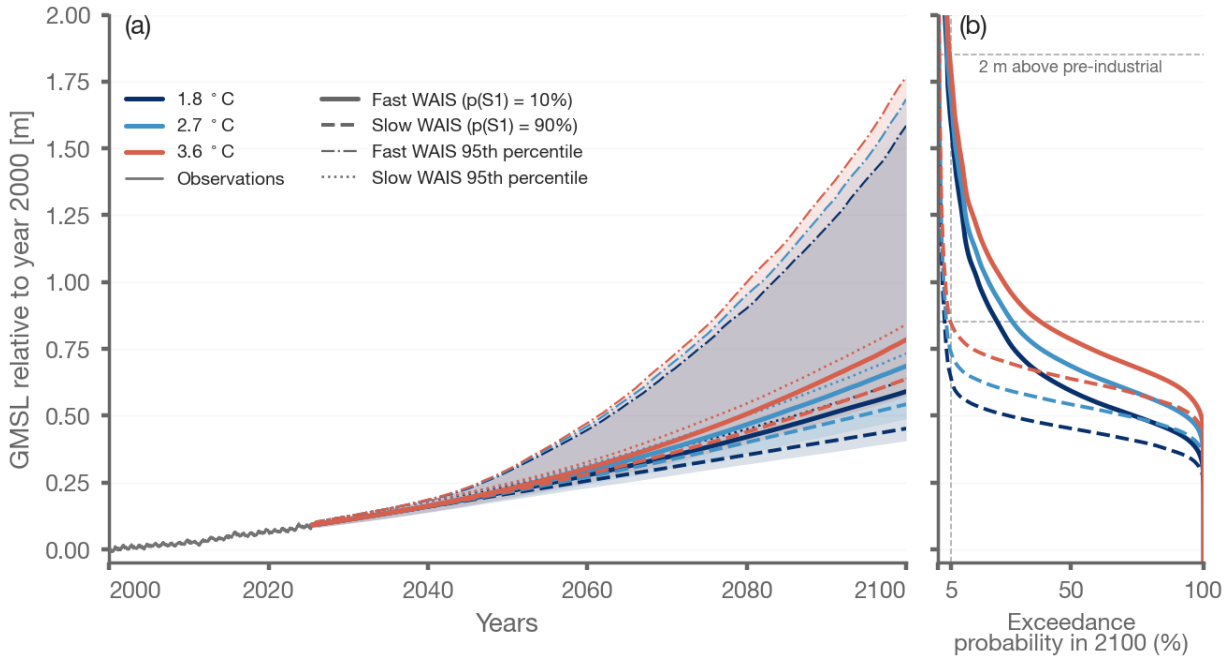


Figure 10: **Global mean sea-level projections.** (a) Global mean sea level relative to 2000 under 1.8, 2.7, and 3.6 °C of warming, with satellite altimetry observations (gray) (*Beckley et al.*, 2017). Solid lines show the median of the fast-WAIS storyline (p(S1) = 10%) with its 90% credible interval (5th–95th percentiles, shaded). Dashed lines show the median of the slow-WAIS storyline (p(S1) = 90%), whose interval is not shaded. Dash-dot and dotted lines mark the 95th percentiles of the fast-WAIS and slow-WAIS forecasts, respectively. (b) Probability of exceeding each sea level in 2100 (horizontal axis) for the fast-WAIS (solid) and slow-WAIS (dashed) forecasts. The vertical axis is shared with (a), so sea level is relative to 2000. Thin gray dashed lines mark 1 m and 2 m above pre-industrial (horizontal) and 5% exceedance (vertical). The 95th-percentile lines in (a) reach the 5% exceedance level in 2100.

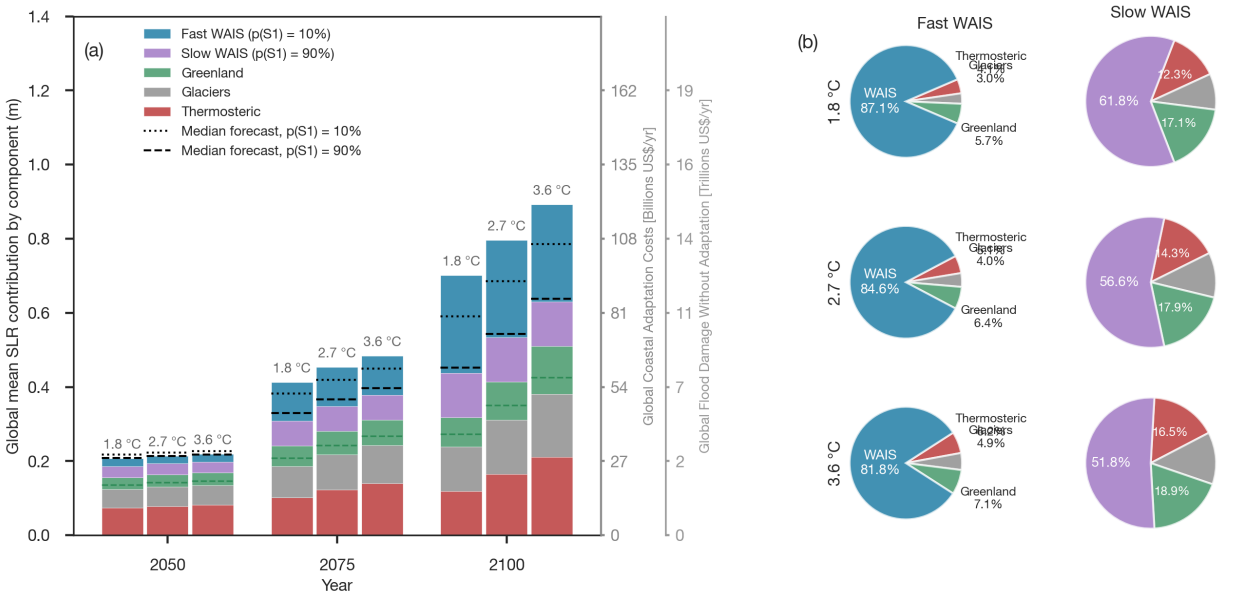


Figure 11: **Component contributions and interval shares.** **(a)** Mean contribution to GMSL rise from the thermosteric, glacier, Greenland, and WAIS components at 2050, 2075, and 2100, and the associated societal costs. Green dashed lines within Greenland bars separate the rise in discharge above its 2000 rate (below) from all other Greenland change (above), which is the SMB change plus the mass imbalance of 2000 carried forward. The purple segment is the mean WAIS contribution under the slow-WAIS storyline, and the blue segment is the additional mean contribution under the fast-WAIS storyline. Black dotted and dashed lines mark the median total forecast under the fast-WAIS and slow-WAIS storylines. The right-hand axes give annual global coastal adaptation costs scaled from *Tiggeloven et al.* (2020), and annual global flood damages without adaptation, interpolated linearly between the values of *Jevrejeva et al.* (2018) (Supplementary Material). **(b)** Shares of the 90% credible interval at 2100, taken as each component's own interval width as a fraction of the summed widths of the thermosteric, glacier, Greenland, and WAIS intervals. The left column shows the fast-WAIS storyline and the right column the slow-WAIS storyline, at 1.8, 2.7, and 3.6 °C from top to bottom.